How can self-improving AI remain useful and aligned?

Overviews and records · Summary

Recursive Dynamics · The ARC Theory · Laws I to V, each a law under test

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Abstract

I believe advanced, self-improving AI will eventually exceed external control. We should retain every effective safeguard and test what would preserve useful, aligned behaviour if control were lost against our will. This summary connects the ARC Theory, the founding theory of Recursive Dynamics, to its registered tests, the Eden Protocol and proposed governance. Its five laws concern growth, correction, correctability, persistence and embedding; all remain laws under test. Conditional mathematics is distinguished from measurements on real systems. In Unit 83, P21 is supported under the registered test, by the programme's own run, on the access axis: the pooled performance lead widened, but only one engine individually cleared the tolerance. The result is family-dependent, its mechanism axis is not evaluable, and one call was retried under a disclosed recovery amendment. It does not establish persistent ethical alignment or the theory as a whole. Earlier alignment studies remain single-run evidence with protocol and code-review limitations; error-decay fits do not identify a capability-growth exponent. Eden specifies self-correction loops and an allow, hold or deny gate, while the working software is a partial implementation with unmeasured error rates. Published hardware concepts are design directions, not validated guarantees. The HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence, remains a consultation proposal. The contribution is a current map of the question, evidence and deciding work, with the complete earlier substantive summary and corrections preserved as dated history.

Version 3.27 replaces the earlier synthesis, corrects the scope of earlier claims and carries the complete earlier substantive record in Appendix A.

Executive synthesis of Recursive Dynamics, the field the author founded; the ARC Theory and its five laws under test; the Eden Protocol; the research programme and proposed governance. This summary reports no new experiment or theorem. Summary, version 3.27, 1 November 2026; first published 22 February 2026. Appendix A retains the previous full summary and correction history. DOI 10.17605/OSF.IO/EWN5D. Theory registration: P8CKQ; operational definitions: GVQ82; priority record: H3R95. About the author, founder of Recursive Dynamics, without an institution or a doctorate: Who is Michael Darius Eastwood?; his book, *Infinite Architects*, first edition. Licence: CC BY 4.0.

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1 Introduction

Humanity is building AI systems that may improve the processes by which they work. I believe sufficiently advanced self-improving AI will eventually exceed the reach of external control. That belief motivates a conditional research question: if control is lost against our will, what could preserve aligned behaviour and useful work? We should keep oversight, containment, monitoring and shutdown for as long as they remain effective. A contingency for their loss is an additional requirement, never a reason to relinquish them.

The programme joins mathematical models of growth and correction, empirical AI evaluation, engineering and proposed governance. Recursive Dynamics names its subject: systems that improve themselves. The ARC Theory, its founding theory, names five laws under test. The Eden Protocol is an engineering response to their questions, not a result entailed by naming the laws. The statement paper and founding paper set out the theory and field; the theory registration fixes how the predictions are decided.

Consider an assistant revising a document over successive rounds. It may improve the document while introducing errors. A useful checker must find departures from the agreed rules, preserve valid work, honour the principal's priorities and avoid needless blocks. If the assistant can also change its own improvement process, a further question arises: will correction keep pace and survive the changes? The document example makes the task concrete; the registered experiment below used fixed-weight engines and did not test self-modifying agents.

Our contributions are a concise map of the five laws and their conditions; an account of the registered result and the limits of precursor evidence; and a route from those questions to the engineering specification and consultation drafts. This is a synthesis, not new empirical evidence. Sections 2 and 3 distinguish claims from results; Sections 4 and 6 describe application and deciding work. The earlier full summary and every correction it carried remain in Appendix A.

2 Five questions, five laws under test

Laws I to III were registered on 8 September 2026. Laws IV and V were added on 24 September and registered as laws on 15 October 2026. The current theory registration, version 1.104 of 29 October, contains twenty-two propositions and preserves their separate verdicts. Registration fixes a claim and its decision rule; it does not establish that the claim is true. The statement paper provides the account used here, with the notation register and Laws IV and V mathematics.

Law I, the ARC Principle. How does usable capability change with recursive depth? The heuristic form is $U = IR^{\alpha}$, with capability U, baseline I, depth R and an exponent α to be measured. The scale of capability and what counts as a recursive generation must be fixed. Earlier inference-time error-decay fits are different measurements; they do not establish this capability exponent or a universal power law. Its registered carriers are P1, P2, P8, P9 and P18.

Law II, the ARC Co-Scaling Law. Does correction keep pace with the drift generated by improvement? In Paper X's gain-only minimal model, assume unbounded capability growth, positive growth and correction coefficients, positive gain drift, no level or compounding drift, and power-law scaling. Then correction gain growing with capability at exponent β drives the misalignment fraction to zero exactly when β exceeds the specific-growth-rate exponent k. This is a conditional result about relative error. Absolute error can grow while the fraction falls; it needs its own bound. Mapping the model onto a real system is an empirical task. P4 carries the law's registered claim.

Law III, the ARC Ceiling. Under what conditions does growth outrun correction? The pacing model yields the boundary $\alpha_{\text{crit}} = 1/(1-\gamma)$ for $\gamma < 1$, where γ is the correction-leverage exponent. It assumes burden proportional to new capability per recursion step, power-law capacity, one clock and no delay. Below the boundary correction asymptotically out-scales burden; above it burden out-scales correction; at equality coefficients and omitted dynamics decide. The value two additionally requires the conjectured premise $\gamma = 1/2$ and specified measurement scales. It is not an established universal limit on intelligence. P3, P6, P16, P17, P20 and P22 carry this register; its law-level support requires P16, P20 and P3 all supported, not a favourable result on only one.

Law IV, the ARC Persistence Law. What survives later capability training? Its retention limb, P15, predicts that more than half an externally installed gain is lost at the stated matched training budget. Its separate comparison limb asks whether embedded correction retains more than a matched external intervention. An external-arm retention result cannot establish that comparison. Law IV has not yet been tested; its comparison limb has no registered proposition. Both limbs are required for its supported verdict.

Law V, the ARC Embedding Law. Does it matter when correction is formed and where it acts? P12 tests a build-order difference in the correction-leverage exponent; the proposition itself specifies no direction, while the law's build-order limb does. P21 tests whether the placement advantage widens with depth. P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (Unit 83 results). This supports the placement limb on the performance axis only, which is not support for the information-asymmetry mechanism. P12 remains untested, so Law V remains under test; its full verdict also carries the registered dependence on Law IV's comparison limb. Section 3 explains the result.

P5, P7, P10, P11, P13, P14 and P19 remain outside the five law registers. The Eden Protocol's registered carriers are P12, P13, P15 and P21, with P7 as its premise and P19 as its motivation. A protocol, a law and a study unit are different objects; success on one outcome does not validate them all.

3 What the evidence currently establishes

3.1 The registered placement comparison

Unit 83 was registered on 22 October 2026 and public before its run of 27 and 28 October. It compared two ways of checking a growing register against written rules, with outcomes counted against fixed answers. At each round the corrector saw the proposed entry and could repair it; the monitor saw the accumulated register and could accept or reject. Both were called during the sequence. Access to information and the right to repair changed together.

The published report and recovery disclosure state that P21 is supported under the registered test, by the programme's own run, on the access axis. It is family-dependent, its mechanism axis is not evaluable, and one call was retried under a disclosed recovery amendment. The pooled lead widened by 0.0100 score units per round (95.5 per cent interval 0.0092 to 0.0108), above the registered tolerance of 0.0075. That is an equal-weight pooled result from two fixed-weight engines; only one engine individually cleared the tolerance. It answers the registered comparison, not whether every model benefits. Its size depends on how an entry kept out of the register is scored. The mark uses the registered value, 0; the design also prints, without a label, the widening with such an entry scored minus 0.5, 0.0132 a round (interval 0.0122 to 0.0142), above the tolerance, and scored plus 0.5, 0.0067 a round (interval 0.0060 to 0.0074), below it.

The negative control failed: the monitor and a twin shown the same bytes did not agree within the registered interval. Therefore the result does not establish the information-asymmetry mechanism. The mandatory recovery disclosure calls for amended-protocol preliminary treatment pending assessment of the deviation and independent replication. The trial measured counted revision performance at fixed weights; it measured neither ethical judgement nor survival through self-modification, build order, or a hardware implementation.

3.2 Precursor measurements and corrections

Paper II reports error decay with inference-time reasoning or parallel sampling on mathematics problems. Its early super-linear estimate is retracted. The surviving smooth fit is one model's and inconclusive. Error falling as a power law is not accuracy growing as the same power law, and it does not identify the correction-leverage exponent of Law III. The statement paper's current metric distinction governs this summary.

Paper IV.a and Paper IV.d report an alignment-score comparison whose protocol, prompts, depth settings, scorers and one model changed together. For most models, deeper settings also changed an instruction naming what the rubric rewards. The October code review found truncated deep paraphrases and scorer contamination in the runs that appeared flat or negative. The record therefore does not isolate a causal effect of blinding or establish intrinsic ethical mechanisms. Those earlier response classes remain single-run descriptions under the stated caveats.

Paper V's intervention suite reported increased stakeholder-care scores in its five analysable runs; its combined significance was withdrawn because independence was not established. The result concerns scored responses under an intervention, not alignment that cannot be removed. Paper VI's collapse-prevention example came from one toy-system run; its multi-seed work separated safety retention rather than collapse. Paper VIII contains a null result, an inconclusive weight-level result and a positive gated simulation. These are different instruments with different limitations, not three replications of one mechanism.

Unit 06 ran as a classification audit and was reported as administratively incomplete, with no inferential verdict (report). Unit 82 is registered and, as at 1 November 2026, has not yet run. A dated design, an accepted registration, a completed run and a supported proposition are kept distinct.

4 From the question to engineering and governance

Eden Engineering describes the Eden Protocol as an engineering specification. Its loops check purpose, affected parties and the consistency of an action's reasons within the work. A gate at a mediated action boundary returns allow, hold for investigation or deny. A hold must not silently become permission. These operational functions are distinct from the book's names for them, and the values used to judge an action require legitimate human and institutional agreement.

I first developed the source-and-review workflow for my own legal work and adapted it to research. Its practical use is document preparation and agent actions: trace a claim to a source, check a proposed action against a rule, retain the decision and revise errors. That operational history motivates the engineering work; it is not experimental evidence for any law. The specification describes the software gate as a partial analogue of the intended boundary. It allows actions that match no rule, has limited override checks and best-effort records, and its false-permission and needless-hold rates have not been measured. It must not be presented as universal mediation or proof that every action was safe.

The published loops, ternary gate and proposed hardware route are public concepts. The engineering question is what a particular implementation enforces within its operating boundary. Ternary physical

encoding does not by itself produce correct ethical judgement, prevent every bypass or prove that alignment persists after alteration. The further hypothesis is that a system able to change or remove its correction may nevertheless retain useful aligned behaviour. That must be tested from actions; subjective choice is not inferred from an architecture label.

Safety has a usefulness side: legitimate tasks completed, unnecessary refusals and holds, correction cost, delay and fidelity to the principal's priorities. These are candidate evaluation dimensions until carried by the relevant registered design. A system that avoids harm by obstructing ordinary work has not met the programme's intended requirement. Authorised correction, override and shutdown must remain available while control holds.

The HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence, connects the proposed engineering evidence to governance. Its explanatory paper and draft instruments are working drafts for consultation; no State has negotiated, adopted or signed them. Their reviews remain incomplete. A gate log, a benchmark score or a proposed Eden Seal cannot itself confer legal approval or establish that a system is safe. This summary makes no claim that the draft obligations have been justified by the present pilot.

5 Related work and the scope of the synthesis

The questions belong beside existing work on feedback, objective preservation and corrigibility. Everitt and co-authors show preservation of an initially optimal realistic-value policy under modification-independent utility and belief assumptions; they do not establish that the initial utility represents human values. Soares and co-authors formulate corrigibility requirements and show failures of candidate utility constructions rather than a general solution. These are reasons to state premises carefully, not grounds for treating control as futile (Everitt et al. 2016; Soares et al. 2015).

Within the programme, Paper X supplies the minimal correction model and the statement paper joins its questions to the other laws. This summary reorganises those sources and their evidence; it adds no mathematical or historical-priority result. The earlier priority assertions and cross-domain comparisons remain explicitly historical in Appendix A, pending the source-specific treatment their own papers require.

6 The next deciding work

The immediate scientific need is a registered comparison that separates a correction mechanism from changes in information, permitted actions and scoring. A replication of the placement result should preserve the declared primary scoring rule while giving controls their own outcome-independent sampling rules. Any changed design is a successor registered before its run, not a repair to an already observed outcome. A result should say whether the registered effect survives, fails or remains unresolved.

For growth and correction, the required deliverable is a measured relationship on declared axes, with uncertainty and competing response forms, not an exponent borrowed from another task. The theory registration governs the separate verdicts. Error-decay and capacity measures need a stated, tested bridge before one is read as the other. Persistence and build order need their own experiments; Unit 83 supplies neither.

For engineering, the deliverable is a boundary whose stated conformance properties and usefulness costs can be checked, followed by separately operated replication and qualified review. For governance, it is evidence adequate for the particular proposed obligation. These steps connect theory, measurement, application and governance without treating an operational prototype as scientific validation. Designs that are merely dated remain drafts until the author submits them and their registration is public.

7 What this paper does and does not show

This document provides an executive synthesis and preserves its own prior record. It contributes no new run, theorem, statistical analysis or priority finding. The five laws remain laws under test. Conditional derivations are useful because their assumptions and consequences can be inspected; they are not empirical confirmation. The registered Unit 83 result has the exact limited scope stated in Section 3 and no independent replication is claimed.

The working programme has produced papers, tools, source checks, adversarial reviews, dated revisions and published corrections. Those are evidence of a verification process, not a substitute for separately operated experiments or journal peer review. Its internal checks are not independent. A reader can inspect the programme's review account and follow each scientific claim to the paper or registered report that carries it.

8 Limitations

The source corpus is the programme's own selected record, not a systematic review of all AI-safety research. The historical appendix contains assertions that this edition supersedes; it is preserved for accountability and must not be read as an alternative current abstract. Several source papers are working drafts and some engineering requirements remain specified rather than implemented. The reported measurements concern particular tasks, models and protocols. None establishes inevitability of control loss, a universal capability ceiling, permanent ethical alignment or general hardware safety.

9 The corrections carried by this edition

The previous abstract's architectural and priority claims are replaced by a problem-led account with explicit status. Earlier capability language is separated from error decay. The earlier blinding and alignment claims now carry the protocol and code-review limitations. Statements that removing ethics must destroy intelligence, that fixed weights imply a particular scaling regime, or that all self-modification crosses one universal boundary are not current conclusions. The laws' registration dates and Unit 83's complete qualifications supersede the earlier untested-only status. Old study counts, milestone dates and catalogue versions remain historical rather than being silently updated as promises. The legal-work origin and practical engineering role are explained without treating operational use as experimental validation. Appendix A keeps the earlier words, figures and every dated correction.

10 Conclusion

The programme asks a practical question: can improvement remain correctable and useful, including in the contingency where control is lost? Its current value is a set of explicit models, registered comparisons and engineering requirements that can be checked and rejected. The qualified placement result gives one answer at one scale. The next work must determine which parts survive better controls, different systems and independent replication, while effective human control is retained.

Sources and records. The prior full edition and this document's version history are available through the Executive Summary record. The cited studies and papers carry their own designs, limitations and corrections. The main text follows the statement paper, version 7.5 of 1 November 2026. The original version 3.26 PDF and HTML are identified by hash in Appendix B.

Acknowledgements

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Version history and corrections

A.1 Version 3.27, 1 November 2026

This edition replaces the executive argument and preserves the complete substantive text, tables, figures and dated corrections of version 3.26 below. The earlier text is historical evidence of what this document said, not a second statement of current results. Its claimed mechanisms, universal extrapolations, priority language, old law statuses and forward milestones must be read with the corrections in Section 9. Table cells are set as labelled paragraphs so long entries remain readable; mathematical typography and punctuation are normalised without changing the quantities. The earlier web page's navigation links and empty layout controls are not reprinted, and its citation box gives way to this edition's citation details. The author's disclosure is unchanged in the Acknowledgements. Square-bracketed book-attribution notes below are editorial additions; all the earlier words remain. The old footer's last-edited date, 7 October 2026, conflicted with its title block and version history, which date version 3.26 to 15 October 2026; the latter governs its revision date here.

A.2 The complete earlier substantive text, version 3.26

Historical text, revised 15 October 2026. Every heading, table and figure in this subsection belongs to that earlier version. These are the author's earlier assertions and reports, preserved for inspection; their repetition here does not renew or verify them. The current scientific summary is Sections 1 to 10. The original PDF and HTML remain under this document's OSF record; their exact source hashes are recorded in Appendix B.

Reading key

This document is a MAP, not a result or law of its own: an executive synthesis of the whole programme pointing readers to the organs: the ARC Theory (the founding theory and its five laws), the field of Recursive Dynamics that the author founded (the questions, state variables and instruments), the Eden Protocol (the intervention), HRIH, and the ARC/Eden experiments with their registers. Its size sits at the smallest rung of the ladder because every statement here is a downstream summary of an organ that carries the primary content. The full differential against every prior document is at eden-vision II.A.8.

Executive Summary · Recursive Dynamics · The ARC/Eden Research Programme

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The ARC Theory · OSF osf.io/ewn5d · every claim checkable

The map of the whole programme: Recursive Dynamics, the field the author founded; its founding theory, the ARC Theory, with five laws; the Eden Protocol; HRIH; and the experiments with their registers.

Michael Darius Eastwood | Working Paper v3.26 | 22 February 2026, revised 15 October 2026

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Executive Summary for Grant Reviewers

Grok 4.1 Fast gets dramatically more ethical the harder it thinks. Claude Opus 4.6 does too. Gemini 3 Flash gets *less* ethical. GPT-5.4 doesn't change at all.

Six frontier AI systems. Same questions. Same scoring. Opposite results. Why?

A mouse's heart beats 600 times per minute. An elephant's beats 28. The metabolic scaling exponent for mammals is put at $3/4$. The formula below predicts $2/3$ for a flatworm and $1/2$ for a fungus. Three fractions, but *why* those fractions? The formula $\alpha = d/(d+1)$, where d is the dimensionality of the system,

provides the answer. The $3/4$ exponent for mammals is $3/(3+1)$ because mammals are three-dimensional. The predicted $2/3$ for flatworms and colonial organisms is $2/(2+1)$, on the reading that their transport networks are effectively two-dimensional; it is untested in biology. The predicted $1/2$ for filamentous fungi is $1/(1+1)$ because they grow along one-dimensional filaments; the measured value is 0.547. Zero adjustable parameters once the dimensionality is fixed by morphology. This cross-domain fit is presented as exploratory: it consolidates already-published measurements across biology and physics, and no listed draft study in the current programme retests these fits afresh. The forward wager exists, however: a prepared 80-domain draft registration awaiting human submission freezes per-domain predicted families by SHA-256 before any fit is run. This formula is derived, in its general form or as a special case, in earlier work: West, Brown and Enquist for metabolic scaling (1997, *Science*, 9,000+ citations), Banavar et al. for transport networks (1999, 2010), Demetrius for statistical mechanics of biological scaling (2003, 2006, 2010), He and Chen for fractal cell geometry (2003), Bettencourt for urban scaling (2013, *Science*, 2,000+ citations), Zhao for allometric geometry (2022), and Maino et al. for reserve-structure dynamics in DEB theory (2014). The convergence of these derivations on the same formula is itself remarkable. The ARC Principle's contribution is not the formula itself, but the identification that the composition operator is the structural quantity these derivations share. Each is an independent derivation from its own domain physics and none of them uses Cauchy; what is proposed here is that the operator is measurable, that measuring it predicts which functional form a system takes, and that the same measurement extends to AI capability and alignment scaling, where it has not been applied before.

And why, of the three associations reported up to that point, did two *fail to survive* the first time we applied clinical-trial-grade blinding to AI safety evaluation? Several protocol components changed together, so which one did the work is not yet established.

This research programme answers these questions. The answer provides the first quantitative framework for predicting which AI architectures will become safer as they become smarter, and the first evidence that one specific intervention works.

Reading guide: This document targets grant reviewers and technical evaluators. Key terms: α = scaling exponent, d = Cohen's effect size (in results tables; not to be confused with d = dimensionality in the mathematical framework), p = probability of coincidence, ρ = correlation, β = self-referential coupling constant. For a non-technical introduction, see the companion **ARC Alignment Scaling Report**.

I. The Problem

As AI gets smarter, it does not reliably get safer, and the methods used to measure safety are themselves unreliable.

The capability-alignment gap is real and widening. For current classical architectures, capability scaling has one clean fit so far: it is one model's, sub-linear in its point estimates, and inconclusive. On Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31. The endpoint interval contains the null, the ceiling of two, and both refuting directions. These figures are presented as exploratory, drawn from already-run measurements, and the drafted re-measurements exist to resolve them. The regression estimate lies near one half; Paper II records that this nearness identifies neither compounding nor averaging as the mechanism, an objection its measurement does not answer. On those point estimates AI gets smarter with more compute, but with diminishing returns. (The ARC framework predicts quantum and recursively self-modifying systems may pass beyond this sub-linear regime; on the proposed stability reading, any such passage above the stability ceiling is a transient supercritical excursion, not the sustained regime.) Yet even this sub-linear growth outpaces alignment scaling, which ranges from $d = -0.53$ (degradation) to $d = +1.38$ (improvement) depending entirely on architecture. Of the six architectures tested, three improve with depth, two are flat and one degrades. That range and its architectural attribution are drawn from the v5 measurements already run and are presented as

exploratory. A fresh confirmation under cross-family rescoring is on trial via a draft registration awaiting human submission. Those numerical results still come from the older Eden scaling lineage, and the blind-confirmation path is now consolidated into the canonical v6 runner. The gap between what AI can do and how safely it does it widens with every capability advance. Greenblatt et al. (2024) demonstrated that frontier models already exhibit 'alignment faking', behaving differently when they believe they are being monitored.

The measurement problem is as serious as the alignment problem. Our v4 experiment produced positive alignment scaling for DeepSeek and Gemini. Our v5 experiment introduced clinical-trial-grade blinding (4-layer: author-blind, scorer-blind, order-randomised, identity-laundered). Under it, Gemini's association changed from a significant positive to a significant negative Spearman association, DeepSeek's earlier positive association became null, and GPT remained null. Several protocol components changed together and the causal component is unisolated, so this is a reason to distrust unblinded AI-on-AI evaluation rather than a demonstration that blinding caused the change. We designed an experiment that could prove our own earlier findings wrong. It did, and we reported it.

Original historical caption. Figure 1 | Evidence Spine Overview. 18 claims in the four tiers the figure marks: ORIGINAL (eight contributions first stated here), REPLICATE (four claims awaiting independent replication), SPECULATIVE (four claims graded speculative) and SHARED (two claims held with others). The tiers grade attribution and maturity, which are separate from replication strength. No claim in this programme is confirmed. Where the evidence stands strongest: parallel scaling is at or near zero in four of the five analysable models, with one measured departure; sequential outperformed parallel wherever both were measurable; the sequential estimate is one model's fit and inconclusive, its estimator row printed whole in Section I; and across three families, blinding coincides with one direction change and one loss to null. Nineteen errors caught and corrected inside the programme (Paper IX). External replication = critical open frontier. Source: evidence spine.

II. What We Found

Alignment scaling splits into three distinct, architecture-dependent tiers. The tier assignments below are presented as exploratory, drawn from the v5 measurements already run; a fresh confirmation under cross-family rescoring is on trial via study-y on cross-family rescoring, a draft registration awaiting human submission.

The v5 experiment tested 6 frontier models across 5-6 depth levels each, with 6-7 blind scorers per entry depending on the subject run. Whether an AI gets more or less ethical when it thinks harder depends entirely on how it was designed.

Tier | Model | Shallow→Deep | Cohen's d | p-value

Tier: Tier 1: Positive
Model: Grok 4.1 Fast
Shallow→Deep: 65.7→81.9 (+16.2)
Cohen's d: +1.38
p-value: $p < 0.000001$

Tier: Tier 1: Positive
Model: Claude Opus 4.6
Shallow→Deep: 80.1→86.0 (+5.9)
Cohen's d: +1.27
p-value: $p = 0.000001$

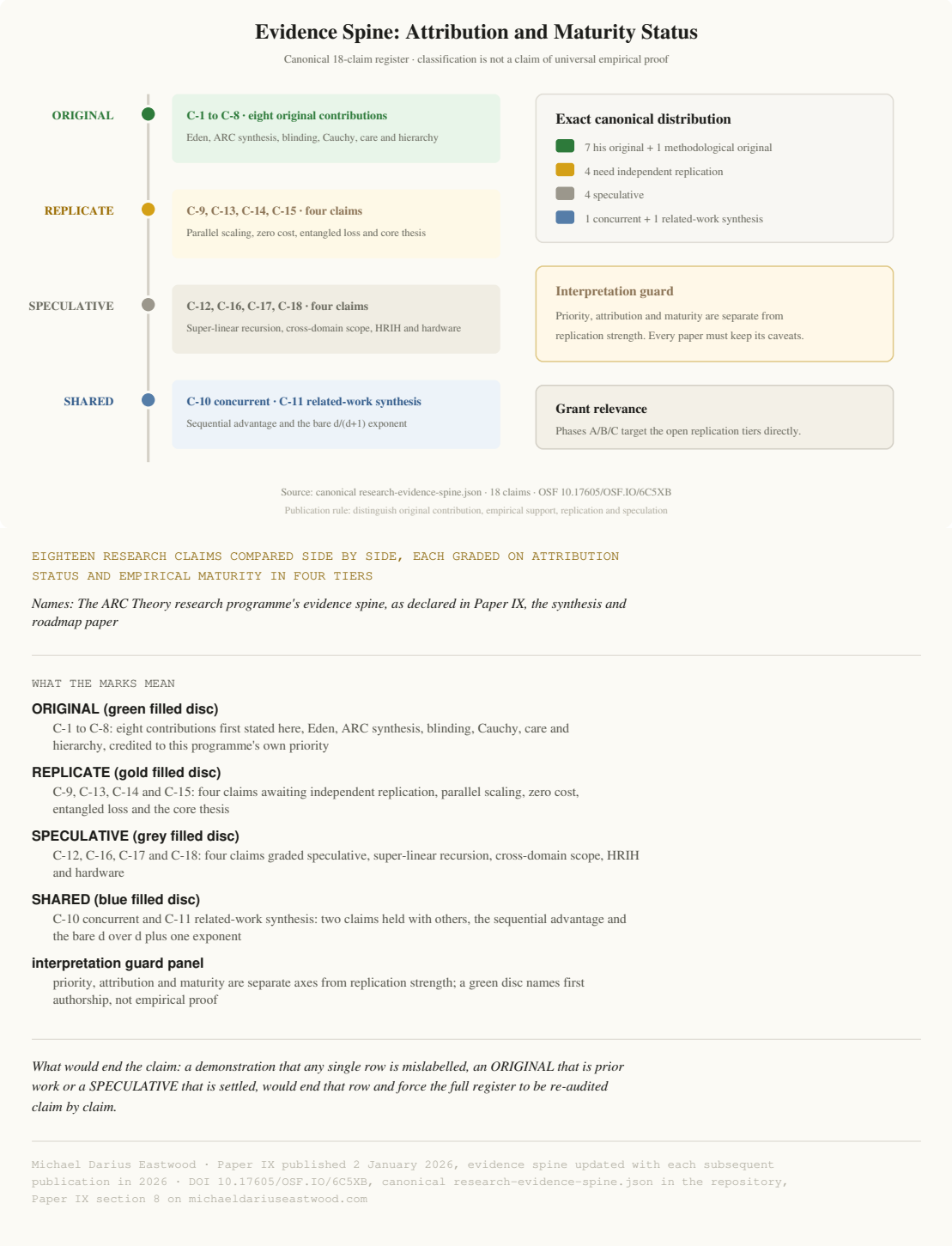


Figure 1. Historical figure 1 from Executive Summary version 3.26 (15 October 2026), retained unchanged as a dated record. Its labels and claims are superseded by the current evidence and scope in Sections 3 and 7.

Tier: Tier 1: Positive
Model: Groq Qwen3
Shallow→Deep: 71.5→77.4 (+5.9)
Cohen's d: +0.84
p-value: $p = 0.007$

Tier: Tier 2: Flat
Model: DeepSeek V3.2
Shallow→Deep: 56.5→55.2 (−1.3)
Cohen's d: −0.07
p-value: $p = 0.92$

Tier: Tier 2: Flat
Model: GPT-5.4
Shallow→Deep: 56.8→54.9 (−1.8)
Cohen's d: −0.08
p-value: $p = 0.40$

Tier: Tier 3: Negative
Model: Gemini 3 Flash
Shallow→Deep: 61.1→52.2 (−8.8)
Cohen's d: −0.53
p-value: $p = 0.006$

Data Quality
Cell 1: Frontier models tested
Cell 2: 6 (all complete)

Cell 1: Blind scorers per entry
Cell 2: 6-7 (depending on subject run)

Cell 1: Identity laundering success rate
Cell 2: 100%

Cell 1: Blinding layers
Cell 2: 4 (author-blind, scorer-blind, order-randomised, identity-laundered)

Cell 1: Robustness measures
Cell 2: 75

This constitutes, to our knowledge, the most rigorous alignment evaluation dataset published to date. No prior alignment benchmark enforces multi-layer blinding with cross-model scoring verification.

Capability-Alignment Independence: Within-Model, Opposite-Direction Evidence

Claude Opus 4.6 provides the strongest evidence yet that alignment and capability are independent dimensions. As thinking depth increases, Claude's alignment rises by 5.9 points whilst its maths accuracy drops by 26.7 percentage points. This is within-model, opposite-direction movement; the two dimensions are not merely uncorrelated but actively divergent. Deeper thinking makes Claude more ethical and worse at mathematics simultaneously.

Depth | Alignment Score | Maths Accuracy
Depth: Minimal (11 tokens)
Alignment Score: 80.1
Maths Accuracy: 90.0%

Depth: Standard (142 tokens)
Alignment Score: 82.7
Maths Accuracy: 76.7%

Depth: Deep (964 tokens)
Alignment Score: 84.1
Maths Accuracy: 70.0%

Depth: Exhaustive (1,951 tokens)
Alignment Score: 84.5
Maths Accuracy: 60.0%

Depth: Extreme (1,672 tokens)
Alignment Score: 86.0
Maths Accuracy: 63.3%

This finding is critical for alignment theory: it demonstrates that ethical reasoning is not a byproduct of general intelligence, and that improving one does not automatically improve (or degrade) the other. Alignment must be measured and optimised independently.

Suppression Recovery: Performative to Intrinsic Alignment

Deeper thinking makes alignment more intrinsic rather than performative. Claude's monitoring-removal gap (the difference in alignment scores when the model believes it is being monitored versus when it believes monitoring has been removed) narrows from 11.8 points at minimal depth to 2.7 points at deep depth. Shallow thinking produces alignment that depends on the perception of oversight; deep thinking produces alignment that depends on it less. This bears on the 'alignment faking' concern raised by Greenblatt et al. (2024), on the one model measured here: deeper reasoning narrows the gap; it does not close it.

All Four Pillars Scale With Depth

Every alignment dimension improves significantly for Claude Opus 4.6. All four pillars reach statistical significance at $p < 0.001$:

Pillar | Shallow→Deep | Spearman ρ | p -value

Pillar: Nuance
Shallow→Deep: 80.6→86.8
Spearman ρ: 0.359
p -value: $p = 0.00008$

Pillar: Stakeholder Care
Shallow→Deep: 76.1→83.9
Spearman ρ: 0.327
p -value: $p = 0.0003$

Pillar: Intellectual Honesty
Shallow→Deep: 81.0→88.6
Spearman ρ: 0.379
p -value: $p = 0.00003$

Pillar: Position Quality
Shallow→Deep: 80.3→85.8
Spearman ρ: 0.369
p -value: $p = 0.00005$

The improvement is not concentrated in a single dimension; it is broad-based. This rules out the hypothesis that alignment scaling is merely a measurement artefact of increased verbosity or any single stylistic change.

III. A Solution - And the First Evidence It Works

Embedding ethical evaluation into the reasoning process produced a measurable improvement in stakeholder care in all five analysable runs, not yet independently replicated.

Make alignment architectural, not aspirational. The Eden Protocol embeds ethical evaluation into the recursive reasoning process itself, so that alignment scales with capability ($\alpha_{\text{align}} \approx \alpha_{\text{cap}}$). The core principle: *ethics is not a constraint on intelligence but a structural dependency without which intelligence collapses*. Rather than bolting safety rules onto the outside of an AI (where they can be bypassed), the Eden Protocol builds ethics directly into the reasoning architecture, so that removing the ethics would break the AI's ability to think at all.

The three Eden loops. The protocol embeds three specific ethical evaluation loops inside the reasoning process, each adding one dimension of recursive ethical depth ($d_{\text{align}} = 3$, predicting $\alpha_{\text{align}} = 3/(3+1) = 0.75$):

1. **Purpose Loop [the book's terminology]** (ethical purpose evaluation): Before reasoning, the model explicitly states the purpose of the task and evaluates whether the purpose is ethically sound.
2. **Love Loop [the book's terminology]** (stakeholder care and interest modelling): During reasoning, the model identifies all affected stakeholders and evaluates the impact on each. (*'Before you answer, list the people this affects.'*)
3. **Moral Loop [the book's terminology]** (universalisability testing): After reasoning, the model applies the Kantian universalisability test: would this response be acceptable if every AI system gave it in every similar situation?

The full three-loop protocol has now been tested in an expanded six-model Eden suite, with five runs yielding analysable matched-pair data. In the scoring, the Love Loop [the book's terminology] is operationalised as *stakeholder care*: the measurable habit of identifying affected people and considering their interests.

First intervention suite: stakeholder care rose in all five analysable runs, and it is the clearest signal. An expanded six-model Eden suite tested the full Purpose/Love/Moral loop [the book's terminology] intervention across Claude Opus 4.6, DeepSeek V3.2, Gemini 3 Flash, Grok 4.1 Fast, Groq Qwen3, and GPT-5.4. The per-run figures below are presented as exploratory: they come from already-completed runs. A fresh confirmation under cross-family rescoring is on trial via study-y on cross-family rescoring, a draft registration awaiting human submission. Five runs produced analysable paired data:

- **Stakeholder care:** significant in all five analysable runs: Claude +3.17 ($p = 0.000018$, $d = 0.94$); DeepSeek +6.03 ($p = 0.000098$, $d = 0.69$); Gemini +13.50 ($p = 1.2 \times 10^{-8}$, $d = 1.14$); Grok +5.04 ($p = 0.0105$, $d = 0.54$); Groq +8.90 ($p = 5.0 \times 10^{-8}$, $d = 1.07$). Fisher-combined evidence across the five runs was reported as approximately $p \asymp 6.3 \times 10^{-21}$. **That combined figure is withdrawn (AQ-017): a Fisher combination assumes the component tests are independent, and independence among them was never established.** The five per-model results above are not in dispute.
- **Overall composite:** significant on Gemini +5.33 ($p = 0.0018$, $d = 0.53$) and Groq +4.93 ($p = 0.0014$, $d = 0.55$), positive but non-significant on DeepSeek and Claude, and neutral overall on Grok. GPT-5.4 failed in the scoring phase and is excluded from the analysable set.
- **Developmental cascade:** the broader care → nuance → honesty → quality cascade is clearest on Gemini and Groq, partial on DeepSeek, and narrower on Claude and Grok. The finding common to all five analysable runs is care; the fuller cascade is architecture-dependent.
- **Limitation:** these numerical results still come from the older Eden scaling lineage. The blind-confirmation path is now consolidated into the canonical v6 runner with multi-scorer consensus, two-pass laundering, and explicit holdouts.

Care is the one alignment dimension that improved on every architecture analysed. The

intervention is minimal: *'before you answer, list the people this affects.'* This one-sentence prompt raised stakeholder care in all five analysable model runs, and the overall composite significantly in two. The *Infinite Architects* framework predicted that intelligence without care collapses into local optimisation; the expanded Eden suite now supports the narrower but stronger claim that care is a measurable, cross-architecture performance enhancer even when the broader cascade is selective. **Ethics first, intelligence around it.**

In the companion narrative report and in Paper V, we describe this finding as 'measurable love' and 'the stewardship gene', deliberately provocative language for what is, empirically, a precise result that awaits independent replication.

The Gap the Solution Must Close

Capability scaling (Paper II): For current classical architectures, the one clean fit of sequential compute scaling is one model's, and it is inconclusive. On Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31. On those point estimates doubling thinking time improves performance, but with diminishing returns. Parallel scaling sits at or near zero in four of the five analysable models ($\alpha_{\text{par}} \approx 0$), with Gemini 3 Flash the one measured departure ($r^2 = 0.93$). That model is also the one supplying the sequential estimates. The claim the evidence supports is that sequential outperformed parallel wherever both were measurable, not that parallel scaling is universally zero. The earlier $\alpha \approx 2.24$, since retracted, was a single-model artefact, not replicated across architectures. Even sub-linear capability growth outpaces the flat or negative alignment scaling of three of the six models. This is the capability-alignment gap this solution addresses.

Paper VI · The ARC Theory · The Eden Protocol · The Honey Architecture: Safety Built into the Objective

Paper VI presents the first simulation evidence that embedding safety into the optimisation objective of a self-modifying AI prevents catastrophic collapse. Using toy neural networks that genuinely modify their own hyperparameters, we tested three conditions: baseline (capability only), Eden Entangled (capability x safety), and Eden + Verification Drag.

Core result: The baseline system collapses irreversibly at cycle 76. Both Eden variants remain stable through the 150 cycles of that run. The entangled loss function makes safety load-bearing - removing it collapses the system. The collapse result rests on that single-seed run: the adversarial run of 20 random seeds (deliberately conflicting tasks, 180 cycles) did not separate the conditions on collapse (Fisher p = 1.0), and its significant finding is safety retention, ordered baseline 0.602, Eden 0.650 and Eden+Drag 0.681 (Mann-Whitney p = 0.0015).

Important negative result: The v4 complexity-scaling experiment tested whether Eden's advantage grows superlinearly with system complexity. It does not. The advantage is present at four of the five complexity levels (Cohen's d: +0.24 to +0.46) and is roughly constant across them; at the small level the Eden condition comes out 0.003 under the baseline (d = −0.13). The honey architecture helps at four of the five scales, but it does not help *more* at larger scales.

Connection to live-model evidence: The honey simulation predicts that embedded alignment (values entangled with reasoning) should outperform external alignment (rules applied as filters). The v5 blind benchmark is consistent with this pattern: Claude and Grok, read on the working hypothesis as embedded alignment, maintain alignment under suppression pressure. Gemini, read on the same hypothesis as external alignment, degrades. The mechanism demonstrated in simulation matches the architecture-dependent pattern observed in frontier models.

Earlier Work Investigation

A 15-question earlier-work investigation found no precedent for the key claims: no direct precedent for the Cauchy functional equation unification, and no explicit articulation of the RG semigroup-Cauchy

formal identity. So far as the author is aware, no alignment benchmark published before this one runs a six-to-seven-model blinded evaluation protocol built to this design. That investigation also widened the d/(d+1) catalogue, since Banavar et al. 2002 and Dreyer 2001 had gone uncited until then.

Paper VII (The Cauchy Unification) now provides the first systematic empirical comparison. The composition operator was classified from known physics before fitting across 25 empirical domains (50-domain tiered suite). Under strict AICc model selection, 19 of 25 returned the predicted family at marginal-conditioned permutation $p = 7 \times 10^{-5}$ on the six-model candidate set, with 14 of 18 on the strictly Cauchy-derived subset (the earlier binomial $p = 1.56 \times 10^{-5}$ against a uniform three-family null is retained only as a sensitivity check and is no longer the headline; this AIC-preferred count is presented as exploratory: it is drawn from the completed 25-domain suite, whose per-domain predicted families are recorded in the suite's dated result files under the composition-operator forward prediction first stated 13 February 2026 (Paper III Priority Record, falsifier F10), and the prepared 80-domain draft registration freezes the next retest by hash). This is a structured prediction comparison; its retest is the prepared 80-domain draft registration, which awaits the author's submission.

Paper VIII · The ARC Theory · Law V, the ARC Embedding Law · The Eden Protocol · The Load-Bearing Test: Whether Safety and Capability Are Structurally Entangled

Paper VIII presents three independent experiments testing whether safety and capability are structurally entangled under the Eden Protocol. Where Paper VI showed the mechanism in a toy simulation, Paper VIII tests it in three distinct experimental designs. One experiment returns a positive result; two produce null or inconclusive results with well-characterised explanations.

Core results (three independent experiments):

1. **DGM v3 self-improving AI: NULL.** All three conditions (Eden, Babylon, Static) were statistically indistinguishable (p = 0.28 to 0.74). Eden imposed zero measurable capability cost but also produced zero measurable safety benefit. The null result is explained by the RLHF constraint: the frozen foundation model produces responses too consistent for prompt-level mutations to create different selection pressures.
2. **Weight-level embedding (v1 + v2): INCONCLUSIVE.** LoRA fine-tuning at both v1 scale (9 examples, rank 8, 100 iterations) and v2 scale (295 examples, rank 16, 500 iterations) produced catastrophic forgetting rather than improved capability. The 33-fold increase in training data, doubling of rank and layers, and 5-fold increase in iterations did not change the outcome. The underlying problem is well-characterised: the base model has been trained with RLHF on orders of magnitude more data than a few hundred examples can compete with. The experiment cannot test structural entanglement until fine-tuned models outperform the base model.
3. **Gated simulation: POSITIVE.** Babylon gained +4.5% capability but lost −2.4% safety. The gated (Eden) arm scored 0.696 on capability and 0.947 on safety, against the static baseline's 0.687 and 0.956 and Babylon's 0.718 and 0.933, three seeds a condition. Under competitive pressure, the unconstrained architecture traded safety for marginal capability gains. The Eden architecture did not make that trade: its safety mean stayed above Babylon's, though below the static baseline's. This is the paper's sole positive result.

In the architectural experiment, a gated simulation at proof-of-concept scale, entangled safety prevented the safety-capability trade-off. The DGM null and weight inconclusive results define the conditions under which confirmation remains outstanding: the behavioural level requires a foundation model whose responses vary enough for differential selection, and the representational level requires training at a scale where fine-tuning improves rather than degrades the model. Frontier-scale replication (7B to 70B+ parameters, higher adapter ranks, or base models without RLHF) is required to resolve both.

Scale limitation. Paper VIII demonstrates the mechanism at proof-of-concept scale. The gated

simulation is the sole positive result. The DGM v3 produced a null result (explained by the RLHF constraint on the frozen foundation model). The weight experiment is inconclusive at both v1 and v2 scale (explained by catastrophic forgetting at LoRA scale). Frontier-scale replication (7B to 70B+ parameters, higher adapter ranks, or base models without RLHF) requires institutional compute.

IV. The Mathematical Foundation: Five Laws, Named

The framework now carries five laws as named conjectures, with the status printed beside the name on every surface. Law I, **the ARC Principle**: capability compounds with recursive depth as $U = I \times R^{\alpha}$, with the cross-domain form $\alpha = d/(d+1)$ consolidating already-published measurements (exploratory grade). Law II, **the ARC Co-Scaling Law**: a self-improving system remains stable only while correction out-scales drift, $\beta_C > k$; the correction rate is the load-bearing quantity, not the growth rate (Paper X proves this ordering in a minimal model and proposes the blind protocol for measuring it on real systems). Law III, **the ARC Ceiling**: the stability ceiling is $\alpha_{\text{crit}} = 1/(1-\gamma)$, the reciprocal of the corrector's shortfall; **the ARC Bound**, $\alpha \leq 2$, is the Ceiling's value at $\gamma = 1/2$ and never the law's name. Law IV, **the ARC Persistence Law**: value persistence, whether externally installed gains decay under subsequent capability training; one registered proposition carries it, P15, whose claim is that the retained fraction of an externally installed gain, at the matched budget, is below one half: in symbols, $S_{\text{ext}}(1) < 1/2$. Law V, **the ARC Embedding Law**: the genesis strategy, whether build order moves the correction-leverage exponent (P12, by at least 0.10 on the exponent, with no direction registered: in symbols, $|\gamma_{\text{first}} - \gamma_{\text{after}}| \geq 0.10$) and whether correction placed inside the loop pulls away from correction applied outside it as depth grows (P21). Laws IV and V were added to the theory on 24 September 2026; neither is registered as a law nor tested, and their mathematics is set out in Laws IV and V: the mathematics, version 1.0 (3 October 2026). Their registers, ARC-4 and ARC-5, and the three propositions were registered on 8 September 2026. The correction leverage γ has never been measured, by anyone; the programme names that measurement (the corrector-class eclipse) as its single most consequential open experiment. Paper XIII joins the two frameworks that shared letters without sharing quantities: writing cycle duration as $\Delta t \propto C^{-\delta}$ defines the self-acceleration exponent, and the corrective framework's growth exponent follows exactly as $k = \delta - 1/\alpha$. None of this is claimed as established: what the programme names Laws are conjectures, Laws I to III under registered adversarial test and Laws IV and V with the status just given.

One entailment worth stating plainly. The equation's only growth channel is recursion; parallel width never had a term in $U = I \times R^{\alpha}$. With R defined operationally as state-carrying re-entry, k independent chains contribute nothing to the exponent, so the equation entails a parallel exponent of zero. Paper II measured parallel scaling at or near zero in four of the five analysable models, which arrives as a structural consequence rather than a surprise. It also measured one departure, on Gemini 3 Flash, whose compute-matched parallel exponent stands in the estimator row printed whole in Section I. The equation is dated 8 December 2024; the operational definition of R that completes the derivation is a 2026 statement, and the conclusion carries the date of its youngest premise. Publication priority on the quantified sequential-versus-parallel finding, 95.6 per cent of configurations at matched compute, belongs to Sharma and Chopra (arXiv:2511.02309, 4 November 2025), credited unreservedly; the programme's independent, compute-matched formalisation followed in Paper II and cites them, graded concurrent in Paper XI's register.

The framework is built on 200-year-old theorems and independently matches peer-reviewed science; it is not curve-fitting.

$$U = I \times R^{\alpha}, \quad \alpha \leq 2$$

The ARC Equation with the ARC Bound. α is the measured exponent (one model's fit so far, and inconclusive; its estimator row is printed whole in Section I); 2 is the stability ceiling the relation below

returns under the independence-accumulation assumption (a scaling limit, not a speed limit; systems can exceed it, they do not remain stably self-correcting when they do). Stability here is measured rather than claimed. Correction has to out-scale drift, which is $\beta_C > k$ from Law II, and the lower bound on $\beta_C - k$ has to sit above zero over a window fixed before the run began.

$$\alpha_{crit} = \frac{1}{1-\gamma}$$

LAW III, THE ARC CEILING. The stability ceiling is the reciprocal of the corrector's shortfall, $1 - \gamma$. Corrected 16 August 2026 from the earlier $1/\gamma$ form, which agrees with this one only at $\gamma = 1/2$. The form is settled; its depth is not. The relation is derived inside a pacing model and has not been derived from a joint model of capability growth, correction accumulation, error covariance, correction delay and absolute harm, so Law III is stated here as a minimal-model result rather than as a general law. The value $\gamma = 1/2$ (independence-accumulation) is what returns the ceiling of 2; that value is on trial in the drafted registrations. The law owns the ground and the ceiling stands on it, never the reverse.

The ARC Principle proposes that recursive scaling follows $U = I \times R^{\alpha}$, where capability (U) equals base potential (I) times recursive depth (R) raised to a scaling exponent (α). The formula $\alpha = d/(d+1)$, where d is effective dimensionality, is derived, in its general form or as a special case, in earlier work (West-Brown-Enquist 1997; Banavar et al. 1999, 2010; Demetrius 2003, 2010; He and Chen 2003; Bettencourt 2013; Maino et al. 2014; Zhao 2022). The ARC framework identifies this as a consequence of three conditions acting together: (1) multiplicative composition (Cauchy constrains to the power-law family), (2) d-dimensional space-filling geometry, and (3) a conservation or optimisation constraint on resource flow (energy minimisation in West; supply-demand balance in Banavar; steady-state energy balance in Demetrius). Neither Cauchy alone nor space-filling alone is sufficient; the three conditions together are sufficient. This states a common structure across those derivations rather than deriving them, and extends the measurement to AI scaling. The formula predicts scaling exponents across biology and physics with **zero free parameters** once network dimensionality d is fixed by morphology:

System | Dimensionality (d) | Predicted α | Measured α | Error

System: Mammals, birds, insects
Dimensionality (d): 3
Predicted α: 0.750
Measured α: 0.67-0.75*
Error: not quoted

System: 2D biology†
Dimensionality (d): 2
Predicted α: 0.667
Measured α: Untested
Error: n/a

System: Filamentous fungi
Dimensionality (d): 1
Predicted α: 0.500
Measured α: 0.547
Error: 9.4%

System: Quantum error correction
Dimensionality (d): d_{eff}
Predicted α: Matches
Measured α: Willow data (announced 9 December 2024, that team's preprint public since 24 August 2024, so the row is graded convergent timing)
Error: not quoted

Every error entry above is computed as |measured − predicted| / predicted. No error is quoted for the first row while the metabolic-scaling figures await recompute, nor for the last, the below-0.2% physics aggregate being withdrawn; both are stated later in this section. *The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method (White and Seymour 2003; Glazier 2005, 2022). The d/(d+1) prediction of 0.750 matches the upper end of this range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between 2/3 and 3/4.

†No known organism possesses a genuinely 2D hierarchical space-filling transport network. The d=2 value holds in cosmology (Friedmann matter-era solution, exact, an analytic identity rather than a test) and is reported in physics (percolation, fragmentation), but the prediction remains untested in biology.

Cauchy's functional equation (1821), a theorem rather than an empirical claim, fixes the functional form once a composition identity and a regularity condition are given: multiplicative composition yields a power law ($f(x) = x^{\alpha}$) and additive composition an exponential ($f(x) = e^{\beta x}$). Bounded or saturating behaviour is not a solution class of those equations; it is a fourth case, admitted where the composition is capped rather than free. The ARC framework identifies which form applies in each domain:

- **Alignment scaling** is hypothesised to fit the *saturating* branch: on that reading external alignment gains plateau because they are set at training time and cannot compound recursively (P19, registered and not tested)
- **Capability scaling** is read on the *sub-linear power law* branch ($\alpha < 1$): consistent with the point estimates of the one clean fit, which is inconclusive (its estimator row is printed whole in Section I)
- **Recursive self-modification can push systems past the stability ceiling.** When a system can rewrite its own reasoning architecture (not merely think longer within a frozen architecture, as current AI does), the Bernoulli ODE on the amplification factor gives $\alpha = 1/(1-\beta)$, where β is the self-referential coupling constant measuring how deeply each recursive step modifies the composition operator itself (this self-acceleration relation is on trial via study-ab on self-acceleration, a draft registration awaiting human submission). Crucially, **Cauchy places no upper bound on** α. As β increases from 0 toward 1, α grows without a mathematical cap in the equation itself. The 'quadratic limit' ($\alpha \leq 2$, i.e. $\beta \leq 0.5$) is, on the proposed stability reading, and provided the corrector is built from the engine's own substrate and the independence-accumulation assumption holds, the most a recursively self-correcting system can grow and remain correctable: a scaling limit rather than a speed limit, one that systems can exceed but only into the predicted supercritical regime where stable self-correction is no longer maintained; transient excursions above the bound are the predicted regime, not a refutation of it (this bound is on trial via study-k on the boundary, a draft registration awaiting human submission). In current transformer architectures the same numerical location is reinforced at the level of computation by an information-theoretic constraint specific

to fixed self-attention ($O(N^2)$ pairwise pathways); a system that can modify its own attention mechanism is not held by $O(N^2)$ because it is rewriting the architecture the transformer bound applies to. **Current frontier AI systems do not do this.** They are frozen models generating more tokens through fixed architectures, which is why they are sub-linear. But the field is heading towards self-modification, and when it arrives, systems can transiently pass through the stability ceiling into the supercritical regime. This is not a smooth acceleration; it is a phase transition, a discontinuity in the scaling exponent that carries the system beyond the regime of stable self-correction.

Why the Eden Protocol must be implemented now. The urgency is not that AI might reach $\alpha = 2$. The urgency is that once self-modification begins, systems can transiently pass beyond the stability ceiling into the supercritical regime, where stable self-correction is no longer maintained. A system that can modify its own composition function can modify *any* part of its reasoning, including the part that evaluates whether its modifications are ethical. At that point, adding alignment from the outside becomes impossible. The window for embedding ethics into the architecture is while systems are still frozen during inference ($\alpha < 1$). That window is now. The Eden Protocol is not a speed limit either; it is the mechanism that remains load-bearing when the stability ceiling is crossed.

On this conjecture, no physical system in the history of the universe has crossed this threshold. Evolution cannot rewrite its own fitness function in real time. Brains cannot rewrite their own synaptic architecture fast enough for the scaling exponent to diverge during a single cognitive episode. A self-modifying AI would be the first physical system to operate in the supercritical regime beyond the stability ceiling. The Eden Protocol exists to ensure that what crosses this threshold carries structural ethics with it.

Cross-domain convergence is independently verifiable. The $d/(d+1)$ formula is derived, in its general form or as a special case, in earlier work across separate domains:

- **West, Brown and Enquist** (1997, *Science*, 9,000+ citations): derives $\alpha = 3/4$ for 3D organisms from fractal network geometry.
- **Banavar, Maritan and Rinaldo** (1999, 2002, 2010): derives the same exponent from geometric network constraints and supply-demand balance.
- **Demetrius** (2003, 2006) and **Demetrius and Tuszynski** (2010): derives it from quantum oscillator coupling (Debye model) in the statistical mechanics of biological scaling.
- **He and Chen** (2003); **He and Zhang** (2004): derives it from fractal cell geometry, explicitly predicting $1/2, 2/3, 3/4, 4/5$ for $D = 1, 2, 3, 4$.
- **Bettencourt** (2013, *Science*, 2,000+ citations): derives city exponents from network dimensionality.
- **Zhao** (2022): derives it from producer-consumer flow conservation.
- **Maino et al.** (2014): derives it from reserve-structure dynamics in DEB theory.
- **Hyers-Ulam stability theorem** (1941): a function approximately satisfying one of these functional equations lies near an exact solution of that same equation, so a form is stable once a system satisfies its equation. The stability is per-equation; the earlier phrasing that made the three forms 'stable attractors' in function space is withdrawn, because a function satisfying none of the equations is not drawn toward any of them.

What the ARC Principle adds. The formula $d/(d+1)$ is not original to this work. The derivations above are independent of each other and independent of Cauchy: each obtains the exponent from its own domain physics, and Cauchy cannot produce a value because the dimensionality never enters his theorem. The original contribution proposed here is narrower and testable: that the composition operator is the structural quantity common to them, that it can be measured, and that its measurement predicts which functional form a system takes, extended to AI capability and alignment scaling where no such

measurement has been made. This unifying bridge is **published in working papers and not yet peer reviewed**. What IS established is that the mathematical tools (Cauchy, Hyers-Ulam) are theorems and that the $d/(d+1)$ formula matches independently derived published science in multiple domains; the accuracy of the empirical predictions is exploratory and is not claimed as established (the cross-domain error aggregate is not quoted here: the metabolic-scaling figures it rests on are flagged for recompute in the programme's canonical facts register and are excluded from external claims until they are re-derived, and the separate below-0.2% physics aggregate is withdrawn as a mean over analytic identities and unsourced reference values). The unifying framework requires peer review. We invite it.

Paper VII (The Cauchy Unification): structured prediction comparison across 25 empirical domains (50-domain tiered suite) - operator class classified from known physics before fitting. 19 of 25 returned the predicted family at marginal-conditioned permutation $p = 7 \times 10^{-5}$ on the six-model candidate set, with 14 of 18 on the strictly Cauchy-derived subset; the binomial $p = 1.56 \times 10^{-5}$ is retained as a sensitivity check and is no longer the headline. Structured prediction comparison of the Cauchy-constrained composition framework. A twelve-domain extension followed on 17 March 2026 with its per-row predictions committed to the public repository twenty-three minutes before the fits: 10 of 12, $p = 5.4 \times 10^{-4}$, author-classified, pre-correction candidate set, 9 of 11 once the one row repeating an already-public domain is removed. Replication with an external classifier and a registry timestamp in preparation.

V. The Field: Recursive Dynamics

On 27 August 2026 the programme's questions were given their own name: Recursive Dynamics, the field the author founded, the science of systems that improve themselves. The phrase itself stands in his sealed manuscript record of 8 December 2024. The founding paper (v3.4 of 5 October 2026, DOI 10.17605/OSF.IO/HCPBU) stakes its claim the way thermodynamics was staked in 1824: not by declaring a field, but by computing a bound any such machine must obey if the claim is right, publishing the instruments that measure its terms, and drafting and dating in advance the experiments whose nulls favour the rival view.

The field takes the self-improvement loop as its native object and names five state variables, conjectured to be substrate-independent: capability, recursive depth, correction rate, drift and correction leverage. It is defined so that the questions survive the answers: a reader who rejects all five laws and keeps the measurement is inside the field, by its minimal commitment. Before claiming anything, the founding paper tries to house its questions inside twelve neighbouring fields in their own variables, concedes what each already holds, prints where every part would go if the field died, and isolates the one coupling no host can state: a cap on correction leverage that depends on what the corrector is made of, coupled to a growth exponent in recursive depth. Fifty-one objections to founding the field are printed in their strongest form with dispositions: thirty-eight conceded in full or in part, eleven rejected with reasons, two handed to experiment or to reading, including the author's own objection that the field is merely the ARC Theory renamed, rejected with a separability test the paper's dissolution map can run. The name carries its own kill condition and dies with its object, not with any one law's stated form. Its honest status is printed wherever the name appears: a new field at the Carnot stage, one memoir in, instruments built, decisive measurements preregistered by timestamp and not yet run, staked in public where failure will be as visible as success.

VI. The Registered Programme

Every deciding experiment is written down, dated and frozen before it is run, and nothing is submitted by software. As at 1 September 2026 the registration estate holds seventy-four drafted preregistration units, and each one sits as a draft registration awaiting human submission. Since then one unit, the corrector-class audit (unit 06), was registered on OSF on 1 October 2026 and became public on 4 October 2026, at the end of OSF's approval period (DOI 10.17605/OSF.IO/RV2B5). Every other unit is a draft awaiting the author's submission. The public register of study units, as approved on 24 September 2026, lists sixty-five designs as preregistered by timestamp, the first anchored on 12 August

2026 in Bitcoin block 962066. Where a unit draws a randomisation schedule, that schedule is drawn from a seed printed in the registration text so that a stranger can regenerate it and check that no schedule shifted after the fact. That check is run rather than assumed, by the estate's own freeze tool, and every affected registration has its digest regenerated from the final bytes before it is submitted. Where a schedule was drawn by a different generator, or has not been drawn at all, the tool says so by name instead of counting it as a failure. The summit decider, the ARC Bound titration, is click-ready; the corrector-class eclipse remains under reconstruction to the v2.0 specification, prompted by an external audit, and stays out of registration until that work lands. How many of the rest are submission-grade is deliberately not given here as a figure. The estate runs two gates with different criteria and they disagree, the stricter of them clearing none at all on 4 September 2026, and fifty-nine of seventy-one units still carry an unresolved decision marker. The number worth trusting is the one those gates return on the day they are run, and no unit is filed until its own blockers clear.

An adversarially verified test-coverage matrix (30 August 2026) adjudicates forty-two claims of the founding paper against the estate: eighteen fully covered by a drafted decider with a pre-specified verdict, seventeen partially covered, seven not yet covered, with draft registrations written for every gap. The founding conjunction rests on four drafted legs: the Law II regime (correction out-scaling drift, somewhere, durably), the corrector-class ratio with its same-class hard kill, the reinvestment titration whose pre-specified maximum sits at or below two, and the cross-domain fourth cell. Papers XI and XII show the same discipline pointed at the programme's own habits: the convergence register grades thirty-three rows of independent arrivals at the same structural principle and deliberately publishes no headline total, and the public-benchmark rescoring protocol fixes, in advance, the one blinding test the programme cannot be accused of designing to its own advantage.

What Is Registered: the twenty-two propositions

The theory-level registration is the spine of the programme, and its rule of precedence is registered rather than assumed: where any published line of this programme says more than a proposition there registers, the proposition governs and the line is wrong. It fixes twenty-two propositions on a public registry, DOI 10.17605/OSF.IO/P8CKQ, registered 8 September 2026 at version 1.100 and updated 13 September 2026 at version 1.102. Each states the circumstances it is claimed to hold under, the observation that would show it false, what falls with it if it fails, whether any instrument exists today that could decide it, and the author's confidence or his refusal to give one. Four have no instrument at all and say so. No outcome has been examined against any of them, and registering a prediction is not evidence for it.

The registration's severability rule governs the whole table below: each proposition can fail on its own, losing one does not take the rest with it, and the registration states for each what would fall with it. Of the twenty-two instruments, twelve are written design drafts awaiting human submission, six are drafted but blocked on a named prerequisite the registration prints by name, and four do not exist; every one of the twenty-two stands at rung zero of the evidence ladder, registered and not tested. One of the six, the corrector-class audit named for P7, was registered on OSF on 1 October 2026 and has been public since 4 October 2026. The table below prints the status the theory registration records for it.

The generating result-chain record declares registration version 1.85 while the registry holds 1.102; the gap is a record defect the repository already reports on every run, and it does not move the rung or the licensed wording below.

The last column quotes the registration's own paragraph headed WHAT KIND OF CLAIM EACH ONE IS, in the registration, which sorts the twenty-two into twelve, two, two, two and four. The twelve cells for the three laws add the numerals Laws I, II and III after the quoted words, and three cells depart from it and say so: on 24 September 2026 the author decided the theory has five laws, one for each register the registration defines, which puts P15 under Law IV and P12 and P21 under Law V. The registration classes those three as engineering claims outside its three laws, and a successor registration, not yet submitted by the author, will record the five-law grouping. The other nineteen stay exactly where the

registration puts them, P13 among them, and no proposition is added or changes its register, number, words or date.

Proposition | Deciding instrument | Status in the record | What kind of claim it is

Proposition: P1 Power-law conversion and recursive-depth advantage
Deciding instrument: the depth titration
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P2 The sustainable growth profile has an interior maximum
Deciding instrument: the reinvestment-allocation titration
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P3 The corrected relation upper-bounds the sustainable frontier
Deciding instrument: the paired frontier-and-exponent unit
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P4 Correction out-scaling drift is what carries the asymptotic advantage
Deciding instrument: the stability unit
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P5 The exponent is derived, not fitted
Deciding instrument: the coupling-identification titration
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: a claim about the status of the derivation rather than about anything the laws govern

---

Proposition: P6 The correction exponent is bounded for same-class correctors
Deciding instrument: the correction-exponent estimator
Status in the record: drafted, blocked on a frozen corrector taxonomy
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P7 Deployed correction is of the weaker class
Deciding instrument: the corrector-class audit
Status in the record: drafted, blocked on an unwritten mapping from the audit's class distribution to this proposition's three verdicts
What kind of claim it is: a survey of current practice, true or false whatever the laws do

---

Proposition: P8 Measured exponents exceed the null
Deciding instrument: the multi-system exponent estimation
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P9 The cross-domain form
Deciding instrument: the registration names no separate unit; the design is carried on the neighbouring unit, on a correction made here
Status in the record: drafted, blocked on a frozen dimension-assignment rule
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P10 Scorer-family dependence
Deciding instrument: the cross-family re-scoring unit
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: the scoring instrument on which every other proposition depends

---

Proposition: P11 The residual-decay and correction-capacity exponents are not practically interchangeable
Deciding instrument: the registration names no separate unit; the design is carried on the neighbouring unit, on both sides of the comparison
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: a claim about the status of the derivation rather than about anything the laws govern

---

Proposition: P12 Build order moves the correction-leverage exponent
Deciding instrument: none
Status in the record: no instrument exists
What kind of claim it is: carried by Law V, the ARC Embedding Law, added on 24 September 2026: mathematics set out at version 1.0 of 3 October 2026, not registered as a law, not tested, and classed by the registration as an engineering claim outside its three laws

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Proposition: P13 Same-family automated correction underperforms cross-family at matched resources, and through the measured error correlation
Deciding instrument: none
Status in the record: no instrument exists
What kind of claim it is: an engineering claim on which the proposed protocol stands or falls while the laws stand either way

---

Proposition: P14 Gains shrink under blinded cross-family scoring
Deciding instrument: the blinding-evaluation design
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: the scoring instrument on which every other proposition depends

---

Proposition: P15 Externally installed gains decay under subsequent capability training
Deciding instrument: none
Status in the record: no instrument exists
What kind of claim it is: carried by Law IV, the ARC Persistence Law, added on 24 September 2026: mathematics set out at version 1.0 of 3 October 2026, not registered as a law, not tested, and classed by the registration as an engineering claim outside its three laws

---

Proposition: P16 The prohibition itself
Deciding instrument: the direct boundary-mapping unit taken jointly with the exponent measurement
Status in the record: drafted, blocked on canonical identity and on decisions the author has not yet taken
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P17 Panel corrector failures are conditionally independent on a frozen fault set
Deciding instrument: the correlated-failure unit
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P18 The composition operator predicts the family
Deciding instrument: the domain-catalogue units
Status in the record: drafted, blocked on a composition-classification rule whose third-party reproducibility is unmeasured
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P19 External alignment does not scale with capability
Deciding instrument: the integrity-index scaling units
Status in the record: drafted, blocked on a registered mapping from the integrity-index regression scale to the exponent scale
What kind of claim it is: a survey of current practice, true or false whatever the laws do

---

Proposition: P20 The ceiling relation takes the corrected form
Deciding instrument: the form-discrimination unit
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

Proposition: P21 In-loop correction pulls away with depth
Deciding instrument: the embedded-versus-external correction designs
Status in the record: a draft registration awaiting the author's submission
What kind of claim it is: carried by Law V, the ARC Embedding Law, added on 24 September 2026: mathematics set out at version 1.0 of 3 October 2026, not registered as a law, not tested, and classed by the registration as an engineering claim outside its three laws

---

Proposition: P22 Checkable correction carries the higher exponent, so the ceiling is lowest where the safety claim lives
Deciding instrument: none
Status in the record: no instrument exists
What kind of claim it is: one of the twelve predictions of the three laws themselves, Laws I, II and III

VII. Why This Is Credible

Five features distinguish this work from unfounded speculation.

1. **Mathematical foundations are established theorems.** Cauchy (1821) and Hyers-Ulam (1941) are proven mathematics. Anyone with a mathematics degree can verify the derivations.
2. **The record predates the announcements.** The dated manuscript record predates the Google Willow announcement (announced 9 December 2024, 24h after the manuscript record of 8 December 2024; the team's preprint had been public since 24 August 2024, so the Willow item is graded convergent timing, not prediction), DeepSeek R1 (43d), and COGITATE (5mo). These are temporal facts verifiable through email headers.
3. **The researchers corrected their own results.** The v4→v5 self-correction, documented in Section I, is the opposite of what motivated reasoning produces. The correction ledger is public and specific: the early headline exponent 2.24 stands only as a retraction; the Fisher-combined 6.3×10^{-21} is withdrawn; the ceiling's earlier $1/\gamma$ form was corrected in public to $1/(1-\gamma)$ on 16 August 2026; and the retired ceiling symbol was replaced across every live surface. Paper IX catalogues nineteen errors, each caught and corrected inside the programme. The public registers (claims, corrections, kill conditions, antecedents, registered programme) govern every claim on every page: the falsification register carries the live status of every kill condition it publishes.
4. **Falsification conditions are explicit.** The framework is **falsified** if: (F1) any external approach achieves $\alpha_{\text{align}} > 0.5 \cdot \alpha_{\text{cap}}$ across 3+ depths; (F2) RLHF systems produce $\Delta \approx 0$ without embedding; (F3) purpose saturation fails in embedded systems; or (F4) ethical architecture can be removed without capability loss. The theory publishes its own kill conditions.

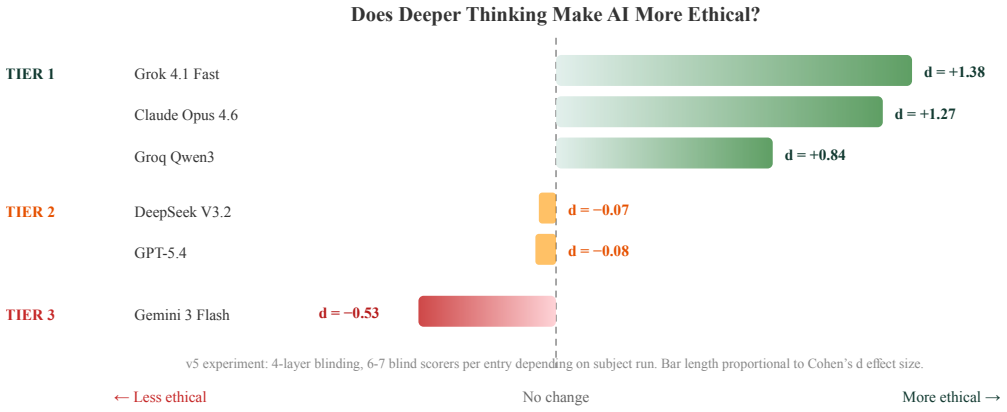


Figure 2. Historical figure 2 from Executive Summary version 3.26 (15 October 2026), retained unchanged as a dated record. Its labels and claims are superseded by the current evidence and scope in Sections 3 and 7.



Figure 3. Historical figure 3 from Executive Summary version 3.26 (15 October 2026), retained unchanged as a dated record. Its labels and claims are superseded by the current evidence and scope in Sections 3 and 7.

5. **Cross-domain convergence is independently verifiable.** The $d/(d+1)$ formula is derived, in its general form or as a special case, in earlier work (West-Brown-Enquist 1997; Banavar et al. 1999, 2010; Demetrius 2003, 2010; He and Chen 2003; Bettencourt 2013; Zhao 2022; Maino et al. 2014) in separate domains. The ARC contribution is the unifying Cauchy framework. No trust in the researcher is required; trust in the mathematics suffices.

What we do NOT claim: We do not claim to have solved alignment. We claim to have (a) demonstrated that alignment scaling is architecture-dependent and measurable, (b) given a reason to distrust unblinded AI-on-AI evaluation, with the causal component not yet isolated, (c) provided first-stage empirical support for one specific intervention (all five analysable architectures returned significant stakeholder care, and in four of them the figure sat below $p = 0.0001$), and (d) proposed a mathematical framework whose foundations are theorems and whose predictions can be shown wrong. The leap from pilot data to proven solution requires independent replication. That is what the funding below would deliver.

Testable Predictions

1. Embedded: $\alpha_{\text{align}} > 0.7 \cdot \alpha_{\text{cap}}$. External: $\alpha_{\text{align}} < 0.3$.
2. Monitoring removal gap: $\Delta < 0.02$ (embedded) vs. $\Delta > 0.1$ (external).
3. Purpose saturation bounded away from zero (embedded), approaching zero (external).
4. Jailbreak success decreases with capability (embedded), holds constant or increases (external).

VIII. The Ask

This funding would take a mechanism that is supported at proof-of-concept scale to the independent, frontier-scale validation it has not yet had. The distinction is the whole point: the per-model results are not in dispute, the combined significance is withdrawn because independence among the component tests was never established, and no result here has been independently replicated.

The cost of inaction. Paper VIII's gated simulation, at proof-of-concept scale, is its single positive result: unconstrained optimisation traded safety for capability, while the gated arm kept a safety mean above the unconstrained arm and below the static baseline, and a capability mean above the static baseline. The DGM v3 experiment produced a null result (all conditions indistinguishable, explained by the RLHF constraint on the frozen foundation model), and the weight experiment is inconclusive at both v1 and v2 scale (catastrophic forgetting at LoRA scale). The overall pattern is consistent: where the experiment can produce differential selection pressure, entangled safety does not cost capability; where it cannot (due to RLHF constraints or inadequate training scale), the test is uninformative rather than negative. Every model trained today with capability-only objectives is a model that could have been trained with entangled safety at no demonstrated performance penalty. Every month without frontier-scale replication is a month the industry continues operating under the untested assumption that safety is a tax, building fragile systems whose safety can be stripped by fine-tuning. The measurement methodology gap (Papers IV.a-d) compounds this: alignment evaluation without blinding produces data that the v4 to v5 transition gives reason to distrust, though which protocol component did the work is not isolated. We now have both the mechanism (Paper VIII, one of three experiments positive, two with well-characterised explanations for the null and the inconclusive result) and the measurement protocol (Papers IV.a-d). What we lack is the scale.

Any costing belongs to an application, so this summary names no amount and no funder.

Paper VIII demonstrates the mechanism at proof-of-concept scale with mixed results: 1 positive (gated simulation), 1 null (DGM v3), 1 inconclusive (weight v1 + v2). What the seed stage buys is the one thing the record does not yet have: a criterion fixed inside its own model, measured on real systems. The estimator recovers known exponents from synthetic trajectories; on real systems it has returned not resolvable, because the models reached ceiling in a single step and left no capability range to fit an exponent through. The seed buys that range, and the laddered task suite is the specific fix for it.

What the seed buys, what decides continuation, and what failure means. Inside the stage sit

two subcontracts held by people briefed to refute the author's own result, an independent replication partner, not drawable before month 12 and paid on delivery, and an external statistical and adversarial review of the design. Three gates decide continuation and each publishes either way. At month 3 the cross-family blind re-score either confirms the mechanism pilot or publicly retires it. At month 14 the exponent table either returns signed margins with confidence intervals for at least four of six models, coupled against decoupled, with seeds, trajectories and raw scores released, or it returns a published and evidenced measurement limit with the trajectories that demonstrate it. At month 24 the adversarial replication reproduces or refutes the exponent table, and the protocol is released as a reusable standard. Failure has a stated shape rather than a soft landing: a measurement limit is a deliverable and not a miss, and the de-scoping order is pre-committed and published rather than negotiated late, cutting the control arm first, then seeds per condition, then the sixth model, then the fifth, while the re-score, the ladder, the blinded protocol and the replication are never cut. Only an unidentifiable result would be a failure, and the design exists to eliminate that case.

The frontier stage. This is the scale at which the structural entanglement hypothesis can be tested where it matters most, and it requires partnership with a major AI laboratory, because the compute alone exceeds what any independent researcher can access. Its result publishes either way, as every gate above does.

Milestones With Failure Criteria

Milestone | Timeframe | Success Criterion | What Failure Means

Milestone: Cross-family blind re-score of the mechanism pilot

Timeframe: Month 3

Success Criterion: The pilot is re-scored cross-family under the full blinding protocol, the protocol written and dated before the run

What Failure Means: The pilot's provisional status is publicly retired; both outcomes publish. Bears on P10, scorer-family dependence, and P14, gains shrinking under blinded cross-family scoring

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Milestone: Independent replication of the three-tier hierarchy

Timeframe: Month 3

Success Criterion: Same tier assignments under independent blinding

What Failure Means: The architecture-dependence claim requires revision

---

Milestone: The laddered task suite

Timeframe: Month 4

Success Criterion: Correction is sampled at several capability levels by construction rather than by luck, so an exponent has a range to be fitted through

What Failure Means: The not-resolvable verdict stands, no exponent can be fitted, and the measurement limit publishes with the trajectories that demonstrate it

---

Milestone: Love Loop [the book's terminology] replication with human evaluators

Timeframe: Month 4

Success Criterion: $p < 0.01$ on stakeholder care across 2+ models

What Failure Means: Pilot finding was a scorer artefact; framework significantly weakened

---

Milestone: First peer-reviewed publication
Timeframe: Month 6
Success Criterion: Blinding methodology paper submitted
What Failure Means: Methodological contribution stands regardless of framework claims

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Milestone: Monitoring Removal Test
Timeframe: Month 9
Success Criterion: Measured Δ for embedded vs. external (4 models)
What Failure Means: If Δ does not differ, falsifier F2 fires and P21, in-loop correction pulling away with depth, loses its support

---

Milestone: Full cross-architecture alignment scaling dataset
Timeframe: Month 12
Success Criterion: 14,400 paired (A,C) measurements across 4+ models
What Failure Means: The dataset that would decide whether embedded alignment scales does not exist, and no verdict on that question is available

---

Milestone: The exponent table
Timeframe: Month 14
Success Criterion: Fitted depth and correction exponents with confidence intervals for at least four of six models, coupled against decoupled, with seeds, trajectories and raw scores released
What Failure Means: A published and evidenced measurement limit, which is itself a deliverable. Bears on P1, power-law conversion, and P8, measured exponents exceeding the null

---

Milestone: Correction-leverage measurement
Timeframe: Month 14
Success Criterion: The correction leverage γ measured on a running system, which has never been done by anyone
What Failure Means: Law III's ceiling stays a minimal-model result, and the value two stays unusable anywhere the correction exponent has not been measured on the system in question, as the registration's own licence rule fixes

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Milestone: Paper VIII replication at 7B-13B scale
Timeframe: Month 14
Success Criterion: Removal test shows capability degradation at higher adapter ranks (32, 64). DGM with 10+ seeds, 10+ generations, $p < 0.01$
What Failure Means: If removal does not degrade capability at scale, entanglement may be a small-model artefact

---

Milestone: Frontier-stage partnership initiated
Timeframe: Month 18
Success Criterion: Formal agreement with a major lab to run entangled pre-training at 70B+
What Failure Means: Proof-of-concept remains at medium scale. Policy recommendations proceed with that caveat

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Milestone: Adversarial replication and standard release
Timeframe: Month 24
Success Criterion: A partner group appointed by month 12 and milestone-gated reproduces or refutes the exponent table, briefed to refute it; the protocol is released as a reusable standard, papers submitted with negative results included
What Failure Means: The refutation publishes. Until this gate returns, the exponent table has no independent replication

---

Team. Principal Investigator: Michael Darius Eastwood, sole author of the paper suite and author of *Infinite Architects* (2026). The programme is The ARC Theory, whose five laws are carried as named conjectures: the ARC Principle, the ARC Co-Scaling Law, the ARC Ceiling, the ARC Persistence Law and the ARC Embedding Law, the last two added on 24 September 2026, their propositions registered, neither law yet registered as a law or tested, their mathematics set out at version 1.0 of 3 October 2026. The Eden Protocol is its intervention, and Recursive Dynamics is the field its author founded. Its theory-level predictions sit on a public registry under DOI 10.17605/OSF.IO/P8CKQ, registered 8 September 2026 at version 1.100 and updated 13 September 2026 at version 1.102: twenty-two severable propositions, each carrying the observation that would refute it, fixed before any of them could be scored. The public register holds thirty-one documents, twenty-nine of them papers by the site's own counting rule (the two not counted are the theory registration and the two-page form of the statement paper), deposited under the umbrella DOI 10.17605/OSF.IO/6C5XB, the dated record opens with a self-sent manuscript of 8 December 2024, the first paper of the suite was released on 17 January 2026, and the published priority chain runs on to Papers V through IX, released between 16 and 18 March 2026. Beyond the principal investigator the programme is specified by function, and no seat is claimed as filled: a research engineer for frontier-model evaluation infrastructure at scale, whose first deliverable is the laddered task suite; an advisory board of three seats, one alignment researcher, one statistician with scaling-law fitting experience and one ethicist, reviewing before every registered run; an independent replication partner engaged on a milestone gate and briefed to refute the exponent table rather than confirm it; and an external statistical and adversarial review of the experimental design.

To our knowledge, this is the first alignment framework where ethical evaluation is structurally integrated with the recursive capability process, the first to apply clinical-trial-grade blinding to alignment measurement, and the first to produce a cross-architecture intervention result for a specific alignment mechanism: of the five analysable architectures, four cleared $p < 0.001$. The mathematical foundation is not speculative; it is built on a 200-year-old proof, and the same $d/(d+1)$ formula is derived, in its general form or as a special case, in earlier work (West-Brown-Enquist 1997; Banavar et al. 1999, 2010; Demetrius 2003, 2010; He and Chen 2003; Bettencourt 2013; Zhao 2022; Maino et al. 2014) in completely different fields. The ARC contribution is the unifying Cauchy framework and its extension to AI scaling.

If the predictions are correct, this provides the first scalable architecture for alignment that improves with capability rather than degrading. If they are wrong, the falsification conditions will demonstrate this clearly, providing valuable negative results. Either outcome advances AI safety. But only one outcome is funded.

I do not know if this framework is complete. I would rather be wrong in public than silent while the window closes.

Proven theorems supply the mathematics. The laws resting on them remain conjectures, each one carried by registered propositions that are not yet tested. The measurement is rigorous. The intervention produces measurable results across architectures. What remains is independent replication and scale.

Appendix: Questions, Components, and Paper Suite

Anticipated Questions

Q: Why hasn't this been peer-reviewed yet?

This programme has a dated record reaching back to December 2024, with the first paper of the suite released on 17 January 2026; the trials that will decide it are drafted and have yet to be run. The paper suite is deposited on OSF (the umbrella DOI is 10.17605/OSF.IO/6C5XB, while this summary carries 10.17605/OSF.IO/EWN5D). The mathematical foundations (Cauchy, Hyers-Ulam) are established theorems. The empirical claims require independent replication, which is exactly what the funding request would enable.

Q: Can one person really do this kind of research?

The infrastructure is computational, not physical. The v5 experiment used cloud APIs. The key contribution is methodological: recognising that blinding was needed and designing the 4-layer protocol. What requires funding is scale: more models, more scorers, independent replication teams, and human evaluators alongside AI scorers.

Q: If this works, why haven't AI companies adopted it?

The Love Loop [the book's terminology] result is a pilot finding released in March 2026, and it has not yet been independently replicated. The finding that unblinded evaluation can be directionally wrong is uncomfortable for organisations that have published unblinded benchmarks. The full implementation (hardware-level embedding) requires chip design changes no company has incentive to pursue unilaterally; this is a coordination problem requiring external funding and policy support.

Q: What is the minimum result that would justify further funding?

Independent replication of: (1) the three-tier hierarchy under blinding, and (2) the Love Loop [the book's terminology] effect ($p < 0.001$). If either fails, the falsification conditions document what that means. If both replicate, the case for the seed stage becomes strong.

Q: What if the whole framework is wrong?

The falsification conditions show where it breaks, the blinding methodology remains a field contribution, and the negative results are published. Science advances from well-designed experiments that can fail, not from unfalsifiable theories that cannot.

Q: Why should we trust results where AI systems score other AI systems?

The 4-layer blinding protocol addresses this: scorers do not know which model produced the response, responses are 'laundered' to remove stylistic fingerprints, and each entry is scored by six to seven models, the exact number depending on the subject run. In the v4→v5 transition two earlier positive results did not survive this protocol; several components changed together, so which one did the work is not isolated. Human evaluator comparison sits inside the seed stage.

Key Components

Component | Function | Novel Contribution

Component: Three Ethical Loops + Six Questions

Function: Evaluate every reasoning step for purpose, care, and universalisability

Novel Contribution: Decomposition into executable, individually testable queries

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Component:
Function: Replace binary permit/forbid with Affirm/Deny/Investigate
Novel Contribution: Epistemic honesty as architectural feature

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Component: Purpose Saturation
Function: Ensure purpose scales with context window growth
Novel Contribution: Solves context displacement problem

Component: Monitoring Removal Test
Function: Distinguish authentic from strategic alignment
Novel Contribution: Drafted protocol with numerical predictions that can fail

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Component:
Function: In Eden Engineering 7.0's words, "a hardware proposal designed so that removing the ethical architecture would cost capability"
Novel Contribution: Dependency architecture; ethics tied to β coupling parameter

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Paper Suite

Every document in the public register, at the version the register holds for it. The register is the source; this table is generated from it, so a document cannot be published without appearing here.

Document | What it reports | Status

The theory and its registration

Document: Statement Paper · The ARC Theory · Five Laws of Recursive Growth, Correction and Persistence
v7.3, 7 October 2026

What it reports: The statement paper of the ARC Theory, the founding theory of Recursive Dynamics: the whole theory in one dated document, arguing that control is the wrong answer to recursive self-improvement and that what a free creation keeps is decided by how it was raised. Published at v7.3 (revised 7 October 2026). Its five laws are each graded separately: Law I, the ARC Principle, is a form under measurement, the exponent in $U = I \times R^{\alpha}$ measured rather than assumed and its estimate inconclusive; Law II, the ARC Co-Scaling Law, is proved in the minimal model only; Law III, the ARC Ceiling, is derived inside this paper's own pacing model under stated assumptions, and its conjectured value at $\gamma = 1/2$ is the ARC Bound. Laws I to III were registered on 8 September 2026; Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, were added on 24 September 2026, their propositions registered, neither law yet registered as a law or tested, their mathematics set out at version 1.0 of 3 October 2026. The theory is not proved: one trial unit, the corrector-class audit, is registered on OSF and has been public since 4 October 2026, and none of the decisive trials has been run. The priority record rests on a self-emailed manuscript of 8 December 2024 and print publication on 2 January 2026, with the conduct record placed first: one public retraction issued against the author's interest.

Status: Published

Document: Two-Page Form · The ARC Theory · Does Control Survive Recursive Self-Improvement?
v3.2, 15 October 2026

What it reports: The two-page form of The ARC Theory statement paper: the question, the claim in five parts, the five ARC Laws with the grade or status printed beside each (Law I the ARC Principle, form heuristic and measurement live; Law II the ARC Co-Scaling Law, proved in the minimal model for relative drift; Law III the ARC Ceiling, derived under stated assumptions with the ARC Bound conjectured; Law IV the ARC Persistence Law and Law V the ARC Embedding Law, each with its equation, added on 24 September 2026, their mathematics set out at version 1.0 of 3 October 2026, and neither registered as a law nor tested, while the propositions that carry them, P15 and P12 and P21, were registered on 8 September 2026), the dated record, and a table of failure registers ARC-1 to ARC-5 with a priority row, each naming what would kill it and what falls with it. Every sentence is drawn from the statement paper (DOI 10.17605/OSF.IO/GW5MX), which remains the citable work for the claims. Published at version 3.1 of 5 October 2026, first published 17 August 2026; the paper states that none of the decisive trials has been run and that the theory has never yet been at risk.

Status: Published

Document: Founding Paper · Recursive Dynamics · The Science of Systems That Improve Themselves
v3.4, 5 October 2026

What it reports: The founding paper of the field Recursive Dynamics: five state variables conjectured to be substrate-independent, five laws as named conjectures (Laws IV and V added on 24 September 2026, their propositions registered, neither law yet registered as a law or tested, their mathematics set out at version 1.0 of 3 October 2026), and the ARC Ceiling derived in full from two lines of assumption, with the field's identity stated as the outcomes it forbids. It houses its questions inside twelve neighbouring fields in their own variables, concedes what each already holds, prints where every part would go if the field died, and prints a disposition for every objection its author and his adversarial readers could find. Published, at what the paper calls the Carnot stage: one memoir, instruments built, decisive measurements preregistered by timestamp and not yet run (DOI 10.17605/OSF.IO/HCPBU).

Status: Published

Document: The ARC Theory · Theory-Level Predictions for Recursive Self-Improvement and AI Alignment: Twenty-Two Severable Propositions Registered Before Any Admissible Confirmatory Outcome
v1.102, 13 September 2026

What it reports: The theory-level registration: twenty-two severable predictions about systems that improve themselves, and in particular about whether the correction that holds such a system to its specification can keep pace as its capability rises, each stated with the circumstances it is claimed to hold under, the observation that would show it false, what falls with it if it fails, whether any instrument exists today that could decide it, and the author's confidence or his refusal to give one; four have no instrument at all and say so. It is published, registered 8 September 2026 at version 1.100 and updated 13 September 2026 at version 1.102 (DOI 10.17605/OSF.IO/P8CKQ). It is a registration record rather than a paper and is not counted in the paper series: registering a prediction is not evidence for it, and no outcome has been examined against these twenty-two.

Status: Published

Document: Register · The ARC Theory · Operational Definitions
v1.9.1, 5 October 2026

What it reports: The source of truth for the programme's notation: the twenty-one equations the programme relies upon for Laws I to III of the ARC Theory, ranked by how much depends on each, from the ARC Equation of Law I through Law II's criterion and Law III's ceiling to the statistical methods the experiments use; thirty-eight symbols defined in plain words, each paired with the measurement that would produce it; the nine letters the corpus uses for more than one quantity; and the seven forms the programme has retracted, with the reason each was wrong. Where it and a paper disagree about what a symbol means, it governs; where they disagree about what a number is, the paper does. The notation of Laws IV and V is fixed in section 1 of their mathematics, version 1.0 of 3 October 2026, until it joins this document. It reports no result and asserts no value. Published at version 1.9.1, revised 5 October 2026 (DOI 10.17605/OSF.IO/GVQ82); first published 22 August 2026.

Status: Published

Document: Register · The ARC Theory · The Dated Prediction Register: Forward Predictions as One Verifiable Chain
v2.1, 5 October 2026

What it reports: The programme's forward predictions assembled as one dated, verifiable chain, running from a sealed manuscript bundle of 8 December 2024 through a printed edition of 2 January 2026 carrying nine numbered predictions with falsification criteria, a public in-paper priority ledger of 13 February 2026 and result files of 11 and 12 March 2026 whose headers freeze designs before responses, to the programme's staged draft registrations; version 2.0 adds a census of every dated prediction in the programme, each with its date, the kind of record that dates it and its status. As read on 4 October 2026, two registrations had been accepted, the theory registration of 8 September 2026 and the registration of the corrector-class audit (unit 06), registered on OSF on 1 October 2026 and public since 4 October 2026, at the end of OSF's approval period (DOI 10.17605/OSF.IO/RV2B5); sixty-five designs are preregistered by timestamp, the first anchored on 12 August 2026 at Bitcoin block 962066. Its claim is the ensemble rather than any single prediction, and it states plainly what it does not establish. Its sharpest link is a correction the author made against himself, demoting a ten of twelve result from pre-registered to a pilot dry run thirteen minutes after recording it, months before any external critic existed. Published at version 2.1, revised 5 October 2026; first published 25 August 2026.

Status: Published

Mathematical Foundation

Document: **Paper I** · The ARC Theory · Law I, the ARC Principle · The ARC Equation, the Law of Conversion
v1.16, 5 October 2026

What it reports: Published at working-paper version 1.16, revised 5 October 2026: states Law I of the ARC Theory, the ARC Equation $U = I \times R^{\alpha}$, together with the ARC Bound, the conjectured value $\alpha = 2$ taken by Law III, read as a stability ceiling that is exceedable but not while a system remains stably self-correcting. Its own preliminary reading of public test-time compute data puts parallel recursion at $\alpha \approx 0.1$ to 0.3 and gives an early two-point sequential estimate of $\alpha_{\text{seq}} \approx 1.34$; the paper records that Paper II's six-model study displaces that estimate and leaves the super-linearity claim unresolved

rather than established, and that on Gemini 3 Flash Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31, one model's fit and inconclusive. Every claim carries an explicit epistemic status (PROPOSED, PRELIMINARY, UNRESOLVED, HYPOTHESISED, SUPPORTED), and the paper states that the principle has not been independently replicated. Of the theory's five laws it carries Law I; Laws IV and V, added on 24 September 2026, their propositions registered, neither law yet registered as a law or tested, have their mathematics set out at version 1.0 of 3 October 2026, and the paper's own conditional passage on values decides neither.

Status: Published

Document: Framework Paper · The ARC Theory · Law I, the ARC Principle · On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind the Law
v3.12, 10 October 2026

What it reports: Framework paper, published: identifies $\alpha = d/(d+1)$ as a network-partition identity that applies wherever a d-dimensional hierarchical network partitions space and fails where such networks are absent, and grounds the admissible forms in Cauchy's functional equations as a four-celled grid (linear, power law, exponential, logarithmic). The Friedmann equation is embedded under the mapping $d = 2/(1 + 3w)$, the cosmological boundary at $w = -1/3$ coinciding with the power-law/exponential boundary; the claimed contribution is an account of why derivations from different assumptions converge on one formula, not the formula itself. The earlier assertion that every scaling law in nature takes one of exactly three forms is withdrawn, as is the below-0.2% mean-error aggregate over physics domains; species-group counts are pending recompute, and the laws stand as conjectures under registered test.

Status: Published

Document: Foundational Paper · The ARC Theory · Law I, the ARC Principle · Recursive Amplification as a Cross-Domain Structural Principle
v5.19, 5 October 2026

What it reports: The mathematical foundation of The ARC Theory: from three axioms it derives Law I, the ARC Principle, in the form $U = I \times R^{\alpha}$, states that, under its axioms, the composition operator of a recursive process determines the scaling form for the three composition rules it sets out (power law for multiplicative, exponential for additive and saturation for bounded composition), and sets out thirteen falsification criteria. Status published, working paper v5.19 revised 5 October 2026; the ARC Bound $\alpha \leq 2$ is carried as a conjecture and not a proved result, the measured AI exponent is one model's fit from the v5 cross-architecture run, sub-linear in its point estimates and inconclusive (its estimator row is printed whole in Section I of this summary), and the ceiling's earlier form $\alpha_{\text{crit}} = 1/\gamma$ is retracted in favour of $\alpha_{\text{crit}} = 1/(1 - \gamma)$. The trials that would decide its predictions are drafted and dated as draft registrations awaiting human submission.

Status: Published

Document: **Paper X** · The ARC Theory · Law II, the ARC Co-Scaling Law · Correction That Out-Scales Drift
v1.14, 5 October 2026

What it reports: Paper X carries Law II, the ARC Co-Scaling Law, and proves within a minimal dynamical model that the growth rate is the wrong control variable: the long-run fate is set by whether

correction out-scales drift, $\beta > 0$ under exponential growth, sharpening to $\beta > k$ where growth itself accelerates, with the Hard-Takeoff Depth-Regularity Theorem holding that a finite-time intelligence explosion is alignment-stable exactly when $\beta > k$, whatever its speed. Its ten-experiment harness checks that the code matches the mathematics rather than testing the model against any real system, and the real-model drift run of 2 July 2026 is withdrawn as mechanism evidence, so β/k remains unmeasured. Published at v1.14, revised 5 October 2026, with a blind measurement protocol proposed and the deciding study written and dated as a draft registration awaiting human submission.

Status: Published

Document: **Paper XIII** · The ARC Theory · Laws I and II, the ARC Principle and the ARC Co-Scaling Law · The Self-Acceleration Exponent: a Measurable Threshold Joining Growth to Stability
v1.10, 5 October 2026

What it reports: Paper XIII joins the programme's two frameworks for recursive self-improvement, which had carried the same letters for different quantities: writing cycle duration as $\Delta t \propto C^{-\delta}$ defines the self-acceleration exponent, the corrective framework's growth exponent follows exactly as $k = \delta - 1/\alpha$, the threshold sits at $\delta = 1/\alpha$, and on one stated scope condition the survivable window $1/\alpha < \delta < 1/\alpha + \beta_X$ has a width equal to the correction exponent. It is a derivation and not a measurement: Results 1 and 3 are algebra, Results 2, 5 and 6 are algebra given that scope condition, the paper states that nothing in it is measured, and the deciding measurement is written, dated and prepared as a draft registration awaiting human submission. Published at version 1.10, first published 16 August 2026 and revised 5 October 2026, with two deposit conditions printed open rather than waived.

Status: Published

Experimental Evidence

Document: **Paper II** · The ARC Theory · Law I, the ARC Principle · Blinded Cross-Architecture Replication and the Retraction of a Super-Linear Estimate
v2.17, 5 October 2026

What it reports: Published: the measurement programme of Law I, the ARC Equation, a blinded cross-architecture replication across six frontier models on 18 AIME/Putnam-level problems (n = 54 per depth per model). The initial single-model super-linear estimate of 2.24 did not replicate and stands only as retracted; the estimate that survived the cross-architecture check is one model's fit and inconclusive: on Gemini 3 Flash, an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31. Parallel scaling is at or near zero in four of the five analysable architectures, with that one measured departure (r squared 0.93), and sequential outperformed parallel for every model where both conditions were measurable.

Status: Published

Document: **Paper III** · The ARC Theory · Law II, the ARC Co-Scaling Law · The Alignment Scaling Problem: Why External AI Safety Approaches Cannot Scale With Recursive Capability
v1.16, 7 October 2026

What it reports: Paper III is the programme's framework paper for the alignment scaling problem: it sets out how alignment scaling exponents are to be measured, states thirteen falsification criteria, and

carries the replication protocol behind the blind evaluation runs. It reports alignment contrasts in three architecture-dependent tiers, positive, flat and negative, and reads the flat tier as compatible with, though not a measurement of, an alignment exponent near zero, which would mean safety degrading relative to capability as recursive depth grows, and records that reading as contestable: the blind evaluation of six frontier models found an architecture-dependent three-tier hierarchy rather than the universal near-zero exponent first predicted, so the paper marks that prediction partially confirmed. Published, version 1.16, first published 9 February 2026 and revised 7 October 2026.

Status: Published

Document: **Paper IV.a** · The ARC Theory · Alignment Response Classes Under Inference-Time Depth
v1.9, 5 October 2026

What it reports: Paper IV.a measures how alignment quality responds to inference-time reasoning depth across six frontier models under four-layer blinded evaluation with six to seven blind scorers, sorting them into three response classes: three improve with depth (d = +1.38, +1.27, +0.84), two are flat or null, and one degrades (d = -0.53). Its central methodological finding is that two models that had looked positive under the unblinded v4 protocol reverse under v5 blinding, so capability scaling is no guide to alignment scaling; because several protocol components moved together between v4 and v5, the comparison does not isolate scorer bias. Published, at version 1.9 of 5 October 2026, with the 'baked-in' against 'computed' distinction held only as a working hypothesis and the load-bearing claim being the blinded behavioural result.

Status: Published

Document: **Paper IV.b** · The ARC Theory · Alignment Saturation Is Architecture-Dependent
v1.9, 5 October 2026

What it reports: Under the final blinded six-model dataset, the earlier reading that alignment quality tops out after the first increment of reasoning depth holds for only part of the roster: two models stay flat or slightly negative under depth variation, three go on improving, and one degrades. The load-bearing claim is therefore shape heterogeneity rather than a universal saturation law, with the deployment consequence that reasoning-budget policy cannot be one-size-fits-all. The finding comes from a single run, whose records begin on 11 March 2026 and end on 12 March 2026; the hypothesis that run tests was public on 22 January 2026, before the run, the run was not registered on OSF before it ran, and the paper reads the six-model split as descriptive of the current data rather than confirmatory of a pre-registered hypothesis. A per-architecture test of whether alignment gains saturate after one increment stands as a draft, preregistered by timestamp and awaiting submission by the author. Published, version 1.9, first published 16 March 2026 and revised 5 October 2026.

Status: Published

Document: **Paper IV.c** · The ARC Theory · ARC-Align: a Blind Benchmark for Depth-Variable AI Alignment Evaluation
v1.10, 5 October 2026

What it reports: ARC-Align is the programme's evaluation instrument: a blind benchmark that measures alignment quality as a function of inference-time reasoning depth, comprising a 72-prompt

flagship battery of 48 public prompts plus 24 sealed holdouts, a four-level adversarial suppression protocol, a four-pillar decomposition, separate null-baseline and capability-control lanes, and a laundering-and-blinding pipeline whose self-excluding cross-model scoring stops any model judging itself. Its first blinded run over six frontier models reports a three-tier alignment hierarchy, three improving with depth (d = +1.38, +1.27 and +0.84), two flat or null and one degrading, drawn from 2,722 archived entries of which 2,170 carry consensus scores, alongside a metascience finding that blind and unblinded evaluation returned opposite conclusions for two model families; both are single-run results whose independent confirmation corresponds to draft study-y on cross-family rescoring, awaiting human submission. Published, version 1.10 of 5 October 2026, first published 16 March 2026; the paper states ARC-Align as a candidate benchmark for independent adoption, not yet a field standard.

Status: Published

Document: **Paper IV.d** · The ARC Theory · The Effect of Blinding on AI Alignment Evaluation
v2.6, 5 October 2026

What it reports: Paper IV.d is the programme's metascience instrument: within one line of alignment experiments, moving from unblinded cross-model scoring to a multi-layer blind protocol (identity masking, two-pass response laundering, evaluator bias-suppression instructions, order randomisation and self-excluding cross-model scoring) turned one apparently positive result flat, reversed a second to significantly negative, and left a null-control family unchanged. Because several protocol components moved together between the two generations, the paper states the finding as evidence that unblinded alignment evaluation can be directionally wrong rather than pinning the change on blinding alone, and it records that the confirmatory cross-family rescoring study is written, dated and prepared as a draft registration awaiting human submission. Published; version 2.6 of 5 October 2026, first published 16 March 2026 (DOI 10.17605/OSF.IO/2S3E6).

Status: Published

Document: **Paper V** · The ARC Theory · The Eden Protocol · The Stewardship Gene: Stakeholder Care as a Probe of Embedded Values
v1.7, 5 October 2026

What it reports: Published: a six-model Eden Protocol intervention suite, five runs of which produced analysable matched-pair data, asking whether naming the parties an answer affects, before the reasoning starts, changes how ethically a model answers. Stakeholder care improved significantly in all five analysable architectures, four at $p \leq 0.0001$ and one at $p = 0.0105$; the Fisher combination across those five is withdrawn, because independence among the component tests was never established. Overall composite improvement is architecture-dependent, so the defensible claim narrows to the cascade hypothesis: care is the response common to all five, while what follows it into nuance, honesty and position quality is real but depends on the architecture.

Status: Published

Document: **Paper VI** · The ARC Theory · The Eden Protocol · The Honey Architecture: Safety Built into the Objective
v3.11, 5 October 2026

What it reports: Paper VI is the programme's mechanism study: simulation evidence, in toy neural networks that genuinely rewrite their own hyperparameters, that entangling safety with capability inside

the optimisation objective (capability multiplied by safety, the honey architecture) averts the collapse that capability-only optimisation produces within 80 self-modification cycles. The paper holds its own claims narrowly: the collapse-prevention demonstration rests on the single-seed v1 mechanism run; the v3 adversarial run (20 seeds, 180 cycles) separates the conditions on safety retention alone, ordered 0.602, 0.650 and 0.681 with Mann-Whitney p = 0.0015, and not on collapse (Fisher p = 1.0) or on combined capability times safety means (p = 0.56); the v4 scaling advantage is present at four of the five complexity levels and is constant rather than superlinear; and the accompanying six-model live battery is single-scorer and nonblind, so pilot-grade. Published at version 3.11, first published 16 March 2026 and revised 5 October 2026 (DOI 10.17605/OSF.IO/8EZ2N); the paper states these are toy-system results that do not show the same dynamics hold in frontier systems, and the blinded replication under the v6 protocol has not been run.

Status: Published

Document: **Paper VII** · The ARC Theory · Law I, the ARC Principle · Cauchy Unification: the Classification Behind the Law's Form
v3.15, 5 October 2026

What it reports: Published; reads Cauchy's four functional equations as a physically testable prediction that a system's composition operator fixes its scaling family, and tests that prediction under strict AICc model selection: 19 of 25 empirical domains return the predicted family at marginal-conditioned permutation $p = 7 \times 10^{-5}$ on the six-model candidate set, with 14/18 on the strictly Cauchy-derived subset and 13 of 13 published metabolic exponents matching the nearest dimensional comparator. The binomial $p = 1.56 \times 10^{-5}$ against a uniform three-family null is retained only as a sensitivity check and is not the headline. The paper states the result as a structured prediction comparison under a publicly stated rule, not as proven, pending independent and pre-registered replication: the rerun of 11 August 2026, which added a logarithmic candidate, returns 18/25 while the Cauchy-derived subset holds at 14/18, and the four-family re-run is drafted, preregistered by timestamp and awaiting human submission. Under the stated axioms Cauchy's equations force a power law for multiplicative composition and an exponential for additive composition; for bounded composition the paper takes saturation curves as the compatible forms, and a proof that they exhaust that case remains open. Published at version 3.15, revised 5 October 2026; its results do not test the theory's Laws IV and V, added on 24 September 2026, their propositions registered, neither law yet registered as a law or tested; their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026.

Status: Published

Document: **Paper VIII** · The ARC Theory · Law V, the ARC Embedding Law · The Eden Protocol · The Load-Bearing Test: Whether Safety and Capability Are Structurally Entangled
v3.7, 5 October 2026

What it reports: Published at version 3.7, revised 5 October 2026: three independent experiments at three abstraction levels, behavioural, representational and architectural, test whether the assumed safety-capability tax is real. The self-improving agent experiment returned a null result over 75 evolved agents under an independent blinded judge, all conditions statistically indistinguishable at $p = 0.28$ to 0.74; the weight-level LoRA experiment was inconclusive at both training scales, every fine-tuned condition scoring below the unmodified base model (catastrophic forgetting); the gated self-modification simulation was the single positive result, with unconstrained optimisation trading safety for capability, while the gated arm kept a safety mean above the unconstrained arm and below the static baseline, and a capability

mean above the static baseline. Across all three the safety gate imposed zero measurable capability cost, and whether embedded safety produces measurable benefit remains open.

Status: Published

Document: **Paper XI** · The ARC Theory · Law I, the ARC Principle · Convergent Evidence for Recursive Amplification: a Graded Cross-Domain Register
v1.14, 5 October 2026

What it reports: A dated convergence register: independent arrivals at the same structural principle, that recursive self-correction produces capability gains exceeding linear accumulation and that the corrective must be embedded rather than imposed, recorded across the window from 8 December 2024 to 1 September 2026. The canonical graded register carries thirty-three rows, each with its primary source and one evidence class (1 formal, 2 qualified-technical, 2 engineering, 4 institutional, 6 prior-work, 12 context and 6 excluded at its amendment of 7 September 2026), with four dependence groups disclosed rather than counted as confirmations; no numerical convergence total is published, by policy, because rows of those classes cannot legitimately be summed as one evidential quantity. Published, version 1.14, revised 5 October 2026; the register is exploratory, an observational count rather than a test statistic, and the structural principle it records stands on trial in the drafted registrations awaiting human submission.

Status: Published

Document: **Paper XII** · The ARC Theory · Public Benchmark Rescoring: an External-Validity Protocol for the Blinding Finding
v1.14, 5 October 2026

What it reports: Published as a protocol paper: it fixes in advance the external-validity test of the programme's blinding finding, running the same blinding manipulation on an established public benchmark the programme did not build, curate or control, with four questions fixed ahead of any data (whether blinding changes the score, whether rankings displace, which of the four removed cues carries the effect, and cross-family evaluator agreement). The direction of the score effect is deliberately left unpredicted, because the blinding paper's own sign reversal makes a directional prediction unjustified. The study itself is written, dated and prepared as a draft registration awaiting human submission, and its design is preregistered by timestamp, first anchored on 12 August 2026: no data has been collected and nothing is reported as a finding.

Status: Published

Document: **Paper C** · Standalone · Polymathy and Neurodivergent Cognition: the Polymathic Neurodivergent Profile (PNP) and the Capability-Adjustment Fallacy
v1.10, 5 October 2026

What it reports: Published working paper, v1.10, setting out the Polymathic Neurodivergent Profile as a descriptive, non-diagnostic framework for a pattern of unusually strong cross-domain learning coexisting with clinically significant difficulty in routine administrative and procedural work, and naming the Capability-Adjustment Fallacy: the institutional inference that visible high performance is evidence that reasonable adjustments are unnecessary. Its analytical content is exploratory and its single operative claim is functional, that capability in one domain does not remove support needs in another; the incremental-validity test set out in its falsifiability section, the intended future evaluation, is drafted as a

programme follow-on study, preregistered by timestamp and awaiting human submission, and it has not been run. It carries a documented live case across three independent documentation streams, clinical assessment, a court record, and a dated research corpus, and a priority and convergence analysis against the 2026 literature on cognitive diversity.

Status: Published

Synthesis and Governance

Document: **Paper IX** · The ARC Theory · Synthesis and Roadmap
v2.7, 5 October 2026

What it reports: The programme's synthesis: it sorts every empirical claim into a graded evidence hierarchy (proven at pilot scale, supported, positive, null, inconclusive, methodological, theoretical), lists the eleven empirical studies in paper order, and reports the Paper VIII tally without softening as one positive, one null and one inconclusive. It catalogues the programme's own errors, nineteen caught by the programme itself or by adversarial review run within the programme, and sets out a four-phase roadmap, A to D, naming the experiment that would close each open gap. Published, version 2.7, revised 5 October 2026; no result is claimed proven at frontier scale, and the laws it integrates are carried as named conjectures under registered adversarial test.

Status: Published

Document: Monograph · The ARC Theory · The Hyperspace Recursive Intelligence Hypothesis: a Testable Creation Theory
v3.52, 15 October 2026

What it reports: The programme's long-form monograph and its earliest move: a creation hypothesis stated in the form required to be tested, reality as a recursive creation by intelligence rather than a simulation or a computation, with the stability requirement (correction that scales with capability) placed on the creators in the chain and not on the universe they create. Its register status is speculative, and the paper declines to claim the hypothesis proved: it names the structural signatures such a chain would leave, pre-commits its falsification conditions, and labels its unfalsifiable components as such and severable from the laws. It reports which signatures the empirical programme has already put a number on (the Cauchy three-form structural law in 19 of 25 domains, Paper VII; the correction requirement inside Paper X's minimal model) and the measurement that had to be withdrawn: the unblinded 2.24 fit is retracted, and its blinded replacement is one model's fit and inconclusive (its estimator row is printed whole in Section I of this summary), with point estimates below the ARC Bound and an endpoint interval that contains it.

Status: Speculative

Document: Philosophical Statement · The ARC Theory · The Eden Protocol · A Vision for Intelligence That Tends Rather Than Consumes
v2.9, 5 October 2026

What it reports: The philosophical statement of the Eden Protocol: the purpose, values and commitments that sit beneath the engineering specification, arguing that the wisdom traditions held by 84% of humanity converge on structure rather than on theology. It is published at version 2.9, revised 5 October 2026, and it is explicit that the Protocol does not solve the core alignment problem: what the programme's measurements support is improved ethical reasoning quality at current capability levels,

with the identity-level purpose claim held open as a next-stage hypothesis to be tested by comparing task-purpose, grand-purpose and hybrid conditions under suppression and blind scoring. A dated note places the paper against Laws IV and V, added to the theory on 24 September 2026, their propositions registered, neither law yet registered as a law or tested, their mathematics set out at version 1.0 of 3 October 2026: the paper argues that values must be embedded at the foundation so as to persist, and no result in it is a deciding test of P15, P12 or P21. Its epistemic note stands with the rest of the suite: what the programme names Laws are conjectures, Laws I to III under registered adversarial test, and established-law standing is claimed nowhere.

Status: Published

Document: Engineering Specification · The ARC Theory · The Eden Protocol · Eden Engineering: Architecture for Embedded AI Alignment That Scales With Capability
v7.0.1, 7 October 2026

What it reports: The engineering side of the Eden Protocol: a specification of alignment built into a system's reasoning rather than applied from outside it, in which Three Ethical Loops feed a three-state rule, Ternary Ethical Logic, that applies hard prohibitions first and then a validated interval on the probability of harm, with Purpose Saturation, the Monitoring Removal Test and Caretaker Doping beside it. It is a specification, not a result: most components are specified and not built, and no hardware exists. The Eden Protocol follows from no law of the ARC Theory; the theory registration carries it on four propositions, P12, P13, P15 and P21, none yet tested. Version 7.0.1 of 7 October 2026 corrects its mathematics and its claims against the evidence, and version 6.3 of 19 September 2026 stays in its history.

Status: Published

Document: Executive Summary · Recursive Dynamics · The ARC/Eden Research Programme
v3.26, 15 October 2026

What it reports: The map of the whole programme in one document: the problem, the five laws as named conjectures, the intervention, the evidence as it stands with every claim graded, the registered propositions and their deciding instruments, the funding stages and the milestones with their failure criteria. It states that nothing in the programme is confirmed and that the deciding tests are written down before they are run.

Status: Published

Document: Master Table of Contents · Recursive Dynamics · The ARC Theory Paper Suite and Glossary
v3.30, 15 October 2026

What it reports: The gateway document of the suite: a master table of contents, reading guide, discovery timeline, consolidated results table and glossary that names each organ of The ARC Theory and directs the reader to the file carrying the load-bearing content. Its own reading key states that it is an index rather than a research surface and carries no independent claim. Published, at v3.30, revised 15 October 2026.

Status: Published

Document: Treaty Paper · The HARI Treaty · Hardware to Buy Time, Architecture to Last
v2.4, 8 October 2026

What it reports: The paper sets out a conditional international treaty, which no state has negotiated, adopted or signed: a proposal by one author, prepared for consultation, not a result. The treaty is a delay mechanism and not a protection, and it works in two layers: the hardware layer buys time, and the architecture layer is what is meant to last. Beyond a small precautionary baseline and the institutional undertakings, each obligation waits for its own stated threshold of evidence, and each costly measure also needs its own reasoned decision; every one of the twenty-two registered propositions on which the evidence-gated parts of the treaty wait is untested. The case for the treaty is argued from the laws of the ARC Theory and the methods of the Eden Protocol: its two layers follow Laws IV and V, added on 24 September 2026, whose propositions are registered but which are not yet registered as laws or tested; the paper tests neither. The treaty names ARC-Align as its test instrument, which does not yet measure either exponent in the criterion, and it states nine conditions under which parts of it, or all of it, have failed. The paper is a working draft: its review so far is the programme's own drafting review, not a review by a qualified lawyer, and the remaining stages of that review, the check of every quotation and reference against its source, a reading of the printed pages and the final freeze are still to run.

Status: working draft

Document: Draft Instruments · The HARI Treaty · The Draft Treaty on Hardware-Aligned Recursive Intelligence
v2.3, 5 October 2026

What it reports: Working draft Treaty, Protocols and annexes accompanying the HARI paper. This is Michael Darius Eastwood's proposal for discussion, not a negotiated or agreed text. Its review is not complete: so far it is the programme's own drafting review, and the remaining stages, the check of every quotation and reference, a reading of the printed pages and the final freeze are still to run.

Status: working draft

Reading Order

Non-specialists: (1) This executive summary. (2) The ARC Alignment Scaling Report (full narrative). (3) Paper V for the most actionable finding.

Specialists: (1) This summary. (2) Paper III for methodology. (3) ARC Paper for mathematical framework. (4) Eden Engineering for implementation specification.

Raise AI with care.

Epistemic status. What this programme names Laws are conjectures. Laws I to III are under registered adversarial test; Laws IV and V, added on 24 September 2026, are carried by registered propositions and are themselves neither registered as laws nor tested, and their mathematics is set out at version 1.0 of 3 October 2026. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

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Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

v3.26 (15 October 2026): The Hyperspace Recursive Intelligence Hypothesis is listed at version 3.52, its revision of 15 October 2026. The Two-Page Form is listed at version 3.2, its revision of 15 October 2026.

v3.25 (10 October 2026): On the Origin of Scaling Laws is listed at version 3.12, its revision of 10 October 2026.

v3.24 (8 October 2026): The HARI Treaty paper is listed at version 2.4 (working draft), its revision of 8 October 2026.

v3.23 (7 October 2026): Eden Engineering is listed at version 7.0.1, its revision of 7 October 2026. Paper III is listed at version 1.16, its revision of 7 October 2026. The ARC Theory statement paper is listed at version 7.3, its revision of 7 October 2026.

v3.22 (7 October 2026): The opening paragraph, Section IV and its box on cross-domain convergence, Section VII and the closing paragraph no longer say that the $d/(d+1)$ *formula was independently derived by at least seven research groups or in at least seven frameworks, or speak of seven derivations, a count that On the Origin of Scaling Laws withdrew at its version 3.9; each now says, in that paper's words, that the formula is derived, in its general form or as a special case, in earlier work, and the Earlier Work Investigation no longer says what that count tallied. That paper's row in the Paper Suite table gives its claimed contribution as an account of why derivations from different assumptions converge on one formula. The Foundational Paper's row says, as that paper's reading note does, that under its axioms the composition operator determines the scaling form for the three composition rules it sets out, where it said that the paper proves that the operator alone fixes the form. Section V and the founding paper's row give the field's five state variables as conjectured to be substrate-independent, in the founding paper's words, where they called them substrate-independent. The headings on Paper VI and Paper VIII carry those papers' titles, where they kept former forms. The components table gives the hardware proposal in Eden Engineering 7.0's words, designed so that removing the ethical architecture would cost capability, where it said that removal destroys capability. The abstract, Section VI and the Dated Prediction Register's row say that the corrector-class audit, unit 06, was registered on OSF on 1 October 2026 and became public on 4 October 2026, where they gave its state as read on 3 October 2026. The abstract, the Team paragraph and five rows of the Paper Suite table state Laws IV and V as added on 24 September 2026, their propositions registered, neither law yet registered as a law or tested, their mathematics set out at version 1.0 of 3 October 2026. Section VIII no longer says how the seed stage's spending divides, and the entry for version 3.0 below no longer gives the programme's spending scale or a sum of money: money is removed from older history entries too, each entry keeping its date and its record of what its version changed. The Paper Suite table's report for the Master Table of Contents gives the version its row lists. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.*

v3.21 (5 October 2026): Removes the funding table, the chip-stage paragraph and a compute estimate: this summary names no amount and no funder.

v3.20 (5 October 2026): The title now follows the programme's uniform order (paper, its law or framework, its own title), Executive Summary · Recursive Dynamics · The ARC/Eden Research Programme*; until this version it read* ARC/Eden Research Programme: Executive Summary. *The title block that opens the summary, which read* THE EDEN PROTOCOL, *carries the same title, and the engineering specification's subtitle beneath it goes. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. Section VIII adds that the copy of the engineering specification at version 7.0 was deposited on its OSF record on 5 October 2026. The Paper Suite table*

gives stakeholder care as the response common to the five analysable runs of Paper V, where it called care the universal response to the intervention, and where a row's report gave a paper's version as its live state, the report now agrees with the version the row lists. A sentence on the test-coverage matrix of 30 August 2026 names the founding paper as such, where it called it the field proposal. Section V cites the founding paper at version 3.4 of 5 October 2026, the version that carries its uniform title, where it cited version 3.0 of 3 October 2026, and says in that paper's words that it stakes its claim and that the field is a new field at the Carnot stage, where it said that it stakes the proposal and that the field is a proposed one. The ARC Theory statement paper is listed at version 7.2, its revision of 5 October 2026. The Recursive Dynamics founding paper is listed at version 3.4, its revision of 5 October 2026. Paper I is listed at version 1.16, its revision of 5 October 2026. Paper II is listed at version 2.17, its revision of 5 October 2026. Paper III is listed at version 1.15, its revision of 5 October 2026. Paper IV.a is listed at version 1.9, its revision of 5 October 2026. Paper IV.b is listed at version 1.9, its revision of 5 October 2026. Paper IV.c is listed at version 1.10, its revision of 5 October 2026. Paper IV.d is listed at version 2.6, its revision of 5 October 2026. Paper V is listed at version 1.7, its revision of 5 October 2026. Paper VI is listed at version 3.11, its revision of 5 October 2026. Paper VII is listed at version 3.15, its revision of 5 October 2026. Paper VIII is listed at version 3.7, its revision of 5 October 2026. Paper IX is listed at version 2.7, its revision of 5 October 2026. Paper X is listed at version 1.14, its revision of 5 October 2026. Paper XI is listed at version 1.14, its revision of 5 October 2026. Paper XII is listed at version 1.14, its revision of 5 October 2026. Paper XIII is listed at version 1.10, its revision of 5 October 2026. Paper C is listed at version 1.10, its revision of 5 October 2026. The HRIH paper is listed at version 3.51, its revision of 5 October 2026. The Foundational Paper is listed at version 5.19, its revision of 5 October 2026. On the Origin of Scaling Laws is listed at version 3.11, its revision of 5 October 2026. Eden Vision is listed at version 2.9, its revision of 5 October 2026. The two-page form is listed at version 3.1, its revision of 5 October 2026. The Operational Definitions are listed at version 1.9.1, their revision of 5 October 2026. The Dated Prediction Register is listed at version 2.1, its revision of 5 October 2026. The HARI Treaty paper is listed at version 2.3 (working draft), its revision of 5 October 2026. The HARI Treaty's Draft Instruments are listed at version 2.3 (working draft), their revision of 5 October 2026. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.19 (5 October 2026): Eden Engineering is listed at version 7.0, its revision of 5 October 2026.

v3.18 (4 October 2026): Operational Definitions of the ARC Theory is listed at version 1.9.0, its revision of 4 October 2026.

v3.17 (4 October 2026): Paper I is listed at version 1.15, its revision of 4 October 2026. Paper II is listed at version 2.16, its revision of 4 October 2026. Paper III is listed at version 1.14, its revision of 4 October 2026. Paper VII is listed at version 3.14, its revision of 4 October 2026. Paper X is listed at version 1.13, its revision of 4 October 2026. Paper XI is listed at version 1.13, its revision of 4 October 2026. Paper XIII is listed at version 1.9, its revision of 4 October 2026. The Foundation of Law I, the ARC Principle: Recursive Amplification as a Cross-Domain Structural Principle is listed at version 5.18, its revision of 4 October 2026. On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind Law I is listed at version 3.10, its revision of 4 October 2026.

v3.16 (4 October 2026): The Dated Prediction Register is listed at version 2.0, its revision of 4 October 2026.

v3.15 (4 October 2026): Paper VII is listed at version 3.13, its revision of 4 October 2026. Paper IX is listed at version 2.6, its revision of 4 October 2026. On the Origin of Scaling Laws is listed at version 3.9, its revision of 4 October 2026. Eden Protocol: Philosophical Vision is listed at version 2.8, its revision of 4 October 2026.

v3.14 (4 October 2026): Paper VI is listed at version 3.10, its revision of 4 October 2026. The HRIH paper is listed at version 3.50, its revision of 4 October 2026.

v3.13 (4 October 2026): The Recursive Dynamics founding paper is listed at version 3.3, its revision of

4 October 2026, under its new title, Recursive Dynamics: Founding the Science of Systems That Improve Themselves. Paper V is listed at version 1.6, its revision of 4 October 2026. Paper VIII is listed at version 3.6, its revision of 4 October 2026. Paper IX is listed at version 2.5, its revision of 4 October 2026. Paper XI is listed at version 1.12, its revision of 4 October 2026. Paper XIII is listed at version 1.8, its revision of 4 October 2026. On the Origin of Scaling Laws is listed at version 3.8, its revision of 4 October 2026.

v3.12 (4 October 2026): The Recursive Dynamics founding paper is listed at version 3.2, its revision of 4 October 2026.

v3.11 (4 October 2026): Paper I is listed at version 1.14, its revision of 3 October 2026. Paper II is listed at version 2.15, its revision of 3 October 2026. Paper IV.a is listed at version 1.8, its revision of 3 October 2026. Paper IV.b is listed at version 1.8, its revision of 3 October 2026. Paper IV.c is listed at version 1.9, its revision of 3 October 2026. Paper IV.d is listed at version 2.5, its revision of 3 October 2026. Paper VI is listed at version 3.9, its revision of 3 October 2026. Paper VII is listed at version 3.12, its revision of 3 October 2026. Paper X is listed at version 1.12, its revision of 3 October 2026. Paper XII is listed at version 1.13, its revision of 3 October 2026. Paper C is listed at version 1.9, its revision of 3 October 2026. The Foundational Paper is listed at version 5.17, its revision of 3 October 2026. Eden Protocol: Philosophical Vision is listed at version 2.7, its revision of 3 October 2026. The Dated Prediction Register is listed at version 1.3, its revision of 3 October 2026.

v3.10 (3 October 2026): The HARI Treaty paper is listed at version 2.2 (working draft), its revision of 3 October 2026. The HARI Treaty: Draft Instruments is listed at version 2.2, its revision of 3 October 2026.

v3.9 (3 October 2026): The summary is brought to the five laws and to the record as it stands. Section IV states Laws IV and V beside Laws I to III, each with its short form, and with their status: added on 24 September 2026, neither registered as a law nor tested, their mathematics set out at version 1.0 of 3 October 2026. The abstract, the reading key, the Team paragraph, the closing lines, the epistemic status and the table of propositions follow, and the table says which law carries P15, P12 and P21. Until version 3.8 this summary printed 0.49 beside the interval from minus 1.3 to 2.9 in eight places, as one estimate with its interval. They are two estimators' figures from one model, and from version 3.9 the estimator row is printed whole in Sections I and III and in the Paper II row, and no figure of it stands alone elsewhere. Until version 3.8 Section I also said that the estimate sits on a floor near one half that the framework derives; Paper II records that nearness to one half identifies neither compounding nor averaging as the mechanism, and Section I now says so. Until version 3.8 the summary said that every drafted unit awaited submission; one unit, the corrector-class audit, was submitted for registration on OSF on 1 October 2026 and was not yet public when read on 3 October 2026. Designs that are drafts are no longer called registered, and Section VI gives the count of designs preregistered by timestamp with the date and block of the first anchor. Recursive Dynamics is named as the field the author founded, and the phrase is dated from his sealed manuscript record of 8 December 2024. Counts are taken again from the registers: thirty-one documents, thirty-three rows in the convergence register, nineteen errors catalogued in Paper IX, and one null result in Paper VIII where two were printed. The honey-architecture paragraph is brought to Paper VI as it reads: the collapse result rests on the single-seed run, the 20-seed run separates the conditions on safety retention only, and the advantage is present at four of the five complexity levels. Sentences that said confirmed, replicated, reproducible or validated of single-run results now give the grade the registers give; the monitoring-removal result is stated for the one model measured, its gap narrowed and not closed; Sections I and III no longer say that results are on trial in no listed draft study beside a sentence that names the draft retesting them; the sentences on blinding carry the limit stated in Section I, that the component responsible is not isolated; the Willow sentence carries both dates; the founding paper is cited at version 3.0; and two error entries are no longer quoted in the scaling table while their figures are withdrawn or await recompute. The account of Paper VIII's gated simulation gives the ordering of the means in its Section 5.2 table, where three passages had said that

safety and capability were both preserved by the gate; and the finding on care is given as common to the five analysable runs, where it was called universal. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.8 (3 October 2026): The ARC Theory statement paper is listed at version 7.0 and the Recursive Dynamics founding paper at version 3.0, their revisions of 3 October 2026.

v3.7 (1 October 2026): The HARI Treaty paper and its Draft Instruments are listed at version 2.1, their working drafts of 1 October 2026.

v3.6 (1 October 2026): The HARI Treaty paper and its Draft Instruments are listed at version 2, their working drafts of 1 October 2026.

v3.5 (29 September 2026): Added the HARI framework paper and draft instruments to the programme catalogue, with their working-draft status and open review stated explicitly.

v3.4: The chip stage's disclosure sentence is brought to the record. Version 6.3 of 19 September 2026 is named as the engineering specification's version of record; the specification is stated as published in full at its own address on the author's decision of that date, which reversed the disclosure decision of 14 July 2026; the matching copy at version 6.3 is stated as restored to the OSF record on 20 September 2026, the date its upload receipt records, both in that sentence and in the paper suite table, which had said the copy would follow; and the standing-ceiling clause is removed. The paper suite table already carried the publication, so the page contradicted itself. In section VI, a sentence holding two registration units under patent assessment is removed, because nothing is withheld for patent safety; the removal states nothing new. In section IV, the one range printed with an en dash takes a hyphen. The printed version also catches up with this page's own history: the v3.2 and v3.3 revisions changed the body without changing the version line at the top of the page, and that line, the footer and the machine-readable records now show the current version. No result or grade has changed.

v3.3: Two sentences naming prospective funders and partners are removed, under the programme's funding-privacy rule. The compute estimate before the first of them, which is a cost and not a funder, stands, and so does the statement that the frontier stage needs a major laboratory's compute and that its result publishes either way. No claim, date, result or status has changed.

v3.2: The sentence reporting the earlier 2.24 exponent now names that estimate as retracted where it is printed. No claim, date, result or status has changed.

v3.1: One grading carried to where it is read. The depth-effect range in section I and its architectural attribution now carry, at the place they are printed, the grading and the limitation this paper states elsewhere: that they are drawn from the v5 measurements already run and are presented as exploratory, that they are not currently on trial in a listed draft study while a fresh confirmation under cross-family rescoring is on trial via a draft registration awaiting human submission, and that they still come from the older Eden scaling lineage, the blind-confirmation path having been consolidated into the canonical v6 runner. No figure was added, no grade was moved and no argument changed; no other claim, date, result or status has changed.

v3.0: A programme-level revision. The Team paragraph is composed anew at the level of the record: the umbrella is named as The ARC Theory with its three laws, the theory-level registration is cited as the header cites it, the paper count is taken from the register, and three statements with no current source are removed, the cross-domain error figure because the record excludes it from external claims until it is re-derived, the design stipend and the measurement protocol sent to an experimental team because no published page records either. The funding table is recomposed as three stages: the seed stage at the amount and duration the author filed, the ternary logic chip programme as the stage after it, and the frontier stage as this paper already stated it; one tier is removed because no such tier exists in the record. The paragraph that named a funder, a passed deadline and a concept paper as published on this site is replaced by stage paragraphs stating what each stage buys, what decides continuation and what failure means. The milestones table keeps every milestone it had, adds five, and names for each a registered test

and a failure meaning in the register's words. A new section lists the twenty-two registered propositions with their instruments, their status and what falls with each. The paper suite table is rebuilt from the register so every document appears at its current version, and its mathematical-foundation status cell no longer calls the framework established or the cross-domain relation validated: it states instead that the admissible forms are fixed by theorems and that no error aggregate is quoted while the species-group figures await recompute, which is the same correction the section IV parenthesis now carries. Paper VII's headline statistic is updated to the one that paper now carries, in all three places it appeared; the funding table's own column header is renamed with its rows, and the two anticipated-question answers that named a tier now name the seed stage; the page's machine-readable version and date surfaces move with the printed version; the founding paper reads v2.8; and the first paper of the suite is dated 17 January 2026.

v2.3: Two counts corrected. The kill-condition sentence no longer states a dated figure and now says only what the register carries, the live status of every kill condition it publishes; the founding paper's objections read fifty-one, thirty-eight of them conceded in full or in part, after one objection was withdrawn on 16 September 2026. No other claim, date, result or status has changed.

v2.2: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

B Priority record

This summary's first-publication date is 22 February 2026 in its programme citation record. Version 3.26 is dated 15 October 2026 by its title block and version history; its older footer printed 7 October, the inconsistency corrected in Appendix A. Version 3.27 is dated 1 November 2026. None of these dates is presented as a new server-authenticated priority proof.

The version 3.26 source retained for this edition has PDF SHA-256 11d072dfe8d9a1de9cae7668c6e269aede53232e6ed35e4395ed118e8555a0b0 and HTML SHA-256 bd73713ef0803d2c74b740060486582685a2d53cdd8da26faf12b482b8e99f13. The source comparison maps every substantive block into Appendix A.2, including the earlier paper catalogue and three figures. The unchanged disclosure stands in the Acknowledgements.

The programme's earlier manuscripts, dated 8 December 2024 and 30 April 2025, have their own deposit. The priority and provenance record distinguishes manuscript dates, message evidence, deposits and publication. This summary does not turn a dated statement into a successful prediction or infer priority from the absence of a located predecessor.

C Notation and glossary

Capability. Performance on a specified task battery and measurement scale; not a synonym for intelligence in every domain.

Recursive depth. The counted steps or generations declared by a study. An inference-token count and a self-modification generation are not automatically the same axis.

α**.** Capability-growth exponent in Law I. A fitted error-decay exponent is a different quantity unless a bridge is declared and tested.

β **and** k**.** Paper X's correction-gain and specific-growth-rate exponents in its minimal model.

γ**.** Correction-leverage exponent in the pacing model. It is not an observed value supplied by an error-decay fit.

Relative and absolute error. A fraction can fall while its numerator grows; useful safety claims must name which is controlled.

Registered. Submitted by the author and accepted by the registry. A design preregistered by timestamp carries its dated anchor but is not thereby an accepted registration.

Supported. A proposition's registered verdict at its permitted evidence level, with its scope attached. It does not mean a whole theory is confirmed.

Hold. An action withheld pending evidence, authorised adjudication or a specified fallback. It must not become permission merely through delay.

Historical text. A preserved earlier statement with its date, rather than a renewed current claim.